Special optimal thermal cells: D-optimal thermal displacements for finite-temperature effective-harmonic phonons

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Finite-temperature effective-harmonic phonons are obtained by fitting force constants to first-principles forces on canonical samples. We introduce special optimal thermal cells (SOTCs), supercell displacements designed from a harmonic trial model in analogy to special quasirandom structures. Cells matching displacement correlators reproduce thermodynamics but do not improve dispersions over canonical sampling. In the D-optimal design SOTC-D, every configuration excites each commensurate mode at its quantum-thermal amplitude, the mode signs maximize the determinant of the force-constant information matrix, and inversion pairs cancel the cubic force. Using the identity of the effective force constants with the Gaussian-averaged Hessian, we show that SOTC-D estimates them with a bias of order $1/M$ in the number of modes M, and that the fixed point of the trial update is the self-consistent harmonic one. For eleven crystals of five structure classes, SOTC-D with eight force calls reproduces the effective dispersions within 0.03–0.31 THz and, with the same trial, has the lowest error of five designs, including canonical sampling, special displacements, and thermal rattle, in 10 of 11 crystals at two and eight calls (two ties). A bootstrapped form updating the trial once outperforms, in the supercell basis, one-shot canonical sampling in all crystals from 4 to 16 calls and two-iteration sTDEP in all crystals at eight calls. On anharmonic model potentials, the self-consistent form needs 2–16 times fewer force calls than the stochastic self-consistent harmonic approximation in 13 of 14 cases. On a volume grid, it gives the thermal expansion of Al within 17% of experiment at 300–400 K.

I. INTRODUCTION

Finite-temperature phonons control phase stability, thermal transport, electron–phonon renormalization, and vibrational thermodynamics[1–3]. At elevated temperature the atomic displacements reach a few percent of the lattice constant, and anharmonic coupling between modes shifts the phonon frequencies at fixed volume.

Finite-temperature lattice-dynamics methods differ in the ensemble over which the forces are averaged and in the anharmonic terms they retain. The quasiharmonic approximation keeps harmonic force constants and lets them depend on volume only, so it omits the explicit temperature dependence that anharmonic coupling produces at fixed volume. Self-consistent harmonic theories include this dependence through an effective harmonic Hamiltonian with temperature-dependent second-order force constants (FC2). In the self-consistent harmonic approximation (SCHA), the FC2 Φ parametrizes a trial harmonic density matrix ρ_Φ; the Gibbs–Bogoliubov upper bound to the free energy is minimized with respect to Φ and the average atomic positions, and at the minimum Φ equals the Hessian of the Born–Oppenheimer surface averaged over the Gaussian displacement distribution of ρ_Φ. The stochastic SCHA (SSCHA) evaluates these averages and the free-energy gradient on configurations drawn from ρ_Φ, and reuses each population over several minimization steps by importance reweighting[4–6]. The temperature-dependent effective potential (TDEP) method is a force-constant fitting procedure rather than a sampling scheme. It takes displacements and forces from a sample of the anharmonic canonical ensemble, most commonly an *ab initio* molecular dynamics (AIMD) trajectory, which treats the nuclei classically[7], and fits a model Hamiltonian of harmonic form, optionally with third- and fourth-order terms, to these forces by least squares[8–10]. Mode-coupling theory provides the formal basis of this construction: the force constants fitted in the canonical ensemble are the static renormalized interactions of the phonon propagator, and the higher-order constants determine its dynamical self-energy[11]. Stochastic TDEP (sTDEP) replaces the AIMD trajectory with configurations drawn from the quantum canonical distribution of the current harmonic model, in which each normal mode has its Bose–Einstein mean-square amplitude. It then repeats fitting and sampling until the force constants are self-consistent[12]. When the average positions are fixed by symmetry, the sTDEP fixed point is the SCHA auxiliary FC2 (Sec. II A). The two methods therefore approach the same effective FC2 by different numerical procedures, which differ in how samples are generated, reused, and weighted. This self-consistent effective FC2 is the target of the present work. It is the SCHA auxiliary FC2, not the physical phonon spectrum of the anharmonic crystal, which in the SCHA follows from the Hessian of the free energy and adds the cubic (bubble) correction[5]; we use "effective dispersion" for the spectrum of the auxiliary FC2 throughout. The SCHA does not require sampling: the self-consistent equations can also be solved with anharmonic force constants computed beforehand, as in self-consistent phonon theory[13]. When the forces are instead taken from first-principles calcula-

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tions on configurations, from a molecular-dynamics trajectory or from Gaussian draws, these are statistical samples of a canonical distribution, and the statistical error of the fitted force constants decreases as the inverse square root of the number of independent samples. The number of first-principles force evaluations needed to converge the force constants is therefore the dominant cost, and it is the subject of this paper.

A deterministic alternative is suggested by the special quasirandom structure (SQS) for substitutional alloys[14]: instead of averaging many disordered configurations, one constructs a single periodic structure whose pair correlations reproduce those of the target ensemble within a cutoff. The special displacement method of Zacharias and Giustino applies a related idea to electron–phonon properties by choosing phonon-mode signs in one supercell[15]. Here we apply the SQS principle to thermal motion for the purpose of effective-FC2 inference. A harmonic trial model defines the equal-time displacement–displacement correlators at temperature T; a special optimal thermal cell (SOTC) is a configuration of a fixed periodic supercell whose shell-averaged correlators match those targets within the force-constant range. First-principles forces on a few such cells are fitted to a constrained FC2 model whose phonons define the next targets.

Whether designed configurations carry more information per force calculation than samples of the canonical distribution is an empirical question. The present paper answers it with controls that hold everything else fixed. For nine benchmark crystals (fcc Al, Ag, Au, Cu; bcc Mo, W; rocksalt NaCl, MgO, PbTe) we (i) analyze the rank, conditioning, and prior sensitivity of correlator-matched SOTC (SOTC-C) fits; (ii) compare the resulting phonons with converged large-cell finite-displacement (FD) references and with an effective FC2 fitted to a 40-snapshot canonical ensemble in the *same* supercell at the same temperature and electronic settings; and (iii) compare SOTC with canonical samples of exactly the same size, fitted by the same regression, on held-out canonical forces, and also with special displacements, thermal rattle, sTDEP, SSCHA (with density-functional theory (DFT) forces for Al and on anharmonic model potentials), TDEP fitted to *ab initio* molecular dynamics (Al), and sparse and Bayesian regression. Diamond Si and hcp Mg are added to test other structure classes. We further report the non-analytic LO–TO correction for the polar crystals, Debye–Waller factors against a tabulated reference, an independent test of constrained third- and fourth-order fits, and the thermal expansion of Al from self-consistent SOTC-D on a volume grid.

This analysis identifies two weaknesses of correlator matching: its scalar objective leaves the amplitude of individual modes, and hence the information content of a configuration, uncontrolled, and its axial regression prior biases the force constants. We therefore introduce SOTC-D, in which each configuration excites every commensurate mode of the trial model at exactly its quantum-thermal r.m.s. amplitude, the mode signs are chosen to maximize the determinant of the FC2 information matrix (D-optimal design), and every configuration is paired with its inverse. An estimator theory (Sec. II) identifies the target as the Gaussian-averaged Hessian, gives the bias of the design exactly, and motivates a self-consistent variant in which the trial is updated during the design. At equal cost, SOTC-D gives lower dispersion errors than the competing designs in most crystals at budgets of two to eight force calls. The gain is largest at the smallest budgets and for crystals whose harmonic trial is far from self-consistency; for a fixed trial it disappears at about 16 calls.

The paper is organized as follows. Section II defines the effective force constants and their self-consistent fixed point, introduces the correlator-matched and D-optimal designs, and derives the bias and variance of the SOTC-D estimator. Section III describes the regression and the iterative workflow, and Sec. IV the first-principles calculations and reference data. Section V compares the designs with each other and with canonical sampling, sTDEP, SSCHA, special displacements, and thermal rattle, for DFT forces and for anharmonic model potentials with known self-consistent solutions. Section VI discusses the scope of the method, and Sec. VII summarizes the results.

II. THEORY

A. Effective harmonic force constants

We consider nuclei moving on the Born–Oppenheimer surface $V(\mathbf{R}+\mathbf{u})$ about fixed reference positions $\mathbf{R}$, with displacements $\mathbf{u}$ and forces $\mathbf{F}(\mathbf{u}) = -\nabla V$. In the self-consistent harmonic description the thermal state at temperature T is represented by a harmonic Hamiltonian with force constants $\mathbf{\Phi}$, whose quantum canonical displacement distribution is a Gaussian $\rho_{\mathbf{\Phi}}(\mathbf{u})$ with zero mean and covariance $\mathbf{C}[\mathbf{\Phi}, T]$ given by Eq. (4) below. For a given covariance $\mathbf{C}$, the effective FC2 is the harmonic model that best reproduces the forces in this ensemble,

$$\mathbf{\Phi}_{\mathrm{eff}}[\mathbf{C}] = \arg\min_{\mathbf{\Phi}} \left\langle |\mathbf{F}(\mathbf{u}) + \mathbf{\Phi}\mathbf{u}|^2 \right\rangle_{\mathbf{C}} = -\langle \mathbf{F}\mathbf{u}^{\mathrm{T}} \rangle_{\mathbf{C}}\, \mathbf{C}^{-1}. \tag{1}$$

For a Gaussian distribution, Stein's lemma[16] gives $\langle \mathbf{F}\mathbf{u}^{\mathrm{T}} \rangle_{\mathbf{C}} = -\langle \nabla\nabla V \rangle_{\mathbf{C}}\, \mathbf{C}$, so that

$$\mathbf{\Phi}_{\mathrm{eff}}[\mathbf{C}] = \langle \nabla\nabla V \rangle_{\mathbf{C}}, \tag{2}$$

the Hessian averaged over the thermal displacement distribution. It contains the renormalization of the harmonic force constants by the even anharmonic terms, at leading order $\frac{1}{2}\sum_b V_{abbe}\sigma_b^2$ in normal coordinates. The self-consistent effective FC2 is the fixed point

$$\mathbf{\Phi}_{\mathrm{SC}} = \mathbf{\Phi}_{\mathrm{eff}}\big[\mathbf{C}[\mathbf{\Phi}_{\mathrm{SC}}, T]\big]. \tag{3}$$

Equation (3) is the stationarity condition of the SCHA free energy with respect to the auxiliary force constants

when the average positions are fixed by symmetry[4,6], and it is the fixed point of sTDEP[12], which alternates sampling from $\rho_{\mathbf{\Phi}}$ and least-squares fitting. TDEP with molecular dynamics evaluates the same least-squares problem with the average taken over the classical anharmonic canonical ensemble instead of a Gaussian[8,9]. One evaluation of Eq. (1) from a harmonic trial gives $\mathbf{\Phi}_{\rm eff}[\mathbf{C}_{\rm harm}]$; we call it the iteration-0 target, and $\mathbf{\Phi}_{\rm SC}$ the self-consistent target. Equations (1)–(3) are standard[4,6,8,12]; the derivation of Eq. (2) from Stein's lemma and its relation to the SCHA free-energy gradient are given in the Supplementary Material, Sec. S4.

In every scheme the average in Eq. (1) is replaced by a sum over a finite set of configurations, each requiring one first-principles force calculation. The estimate is a linear regression of forces on displacements, and its accuracy for a given number of force calculations is determined by how the configurations are chosen. Canonical (Gaussian) sampling, as in sTDEP and SSCHA, draws them independently from $\rho_{\mathbf{\Phi}}$. The rest of this section describes two deterministic alternatives and the statistical properties that distinguish them.

B. Thermal displacement correlators

For a crystal with n_b basis atoms and N_c primitive cells, the equal-time canonical correlator of a harmonic lattice is[17]

$$C_{ab}^{\alpha\beta}(\mathbf{R},T) = \frac{\hbar}{2N_c\sqrt{M_aM_b}} \sum_{\mathbf{q}s} \frac{e_a^{\alpha}(\mathbf{q}s)e_b^{\beta *}(\mathbf{q}s)}{\omega_{\mathbf{q}s}} \times \coth\frac{\hbar\omega_{\mathbf{q}s}}{2k_BT}\, e^{-i\mathbf{q}\cdot\mathbf{R}}, \tag{4}$$

where $\mathbf{q}$ runs over the wave vectors commensurate with the Born–von Kármán cell, s labels branches, and e are mass-weighted polarization vectors (standard result[17]; derivation in Supplementary Material, Sec. S10). The coth factor interpolates between zero-point motion and classical equipartition. The SOTC target is the shell- and trace-averaged scalar

$$\bar{C}_2^{ab}(R,T) = \frac{1}{3N_R^{ab}} \sum_{\mathbf{R}'\in\mathcal{S}_R^{ab}} \sum_{\alpha} C_{ab}^{\alpha\alpha}(\mathbf{R}',T), \tag{5}$$

where $\mathcal{S}_R^{ab}$ is the neighbor shell of radius R for the ordered basis pair (a,b), with multiplicity N_R^{ab}. Shells are formed by binning pair distances at 0.05 Å separately for each species pair. The on-site value $\bar{C}_2^{aa}(0,T)$ is the isotropic displacement parameter $U_{\rm iso} = \langle|\mathbf{u}_a|^2\rangle/3$, related to the crystallographic Debye–Waller factor by $B = 8\pi^2 U_{\rm iso}$.

C. Correlator-matched cells (SOTC-C)

For a candidate displacement field $\mathbf{u}$ of an n-atom supercell, the SOTC objective is

$$\mathcal{Q}(\mathbf{u}) = \sum_{a,b} \sum_{R\le r_{\max}} e^{-R/\xi_w} \left[\bar{C}_{2,ab}^{\rm cell}(R;\mathbf{u}) - \bar{C}_{2,ab}(R,T)\right]^2, \tag{6}$$

where $\bar{C}_{2,ab}^{\rm cell}$ is evaluated from the single configuration with the same shell, trace, and species averaging as Eq. (5), and ξ_w is a numerical weighting length ($\xi_w = r_{\rm cut}/2$; the gradient is given in Supplementary Material, Sec. S11). The optimization starts from a superposition of commensurate trial phonon modes with harmonic amplitudes $A_{\mathbf{q}s} = [\hbar\coth(\hbar\omega_{\mathbf{q}s}/2k_BT)/2\omega_{\mathbf{q}s}]^{1/2}$ and random phases, and adjusts mode amplitudes and phases by gradient descent with restarts; rigid translations are removed. Each iteration produces $N_{\rm ens}$ such configurations with different random starts. The objective constrains only the selected scalar moments: it does not by itself determine the full tensor covariance, and the matched controls below are needed to assess its value for force-constant inference. We refer to cells designed with this objective as SOTC-C.

D. D-optimal thermal-mode design (SOTC-D)

Equation (6) constrains shell sums of second moments, but not the amplitude with which individual modes appear in a configuration; a mode that happens to be weakly excited contributes little information to the force fit. SOTC-D removes this freedom. Let ω_m and $\mathbf{e}_m$ be the commensurate normal modes of the trial FC2 in the supercell (translations excluded) and $\sigma_m^2 = \hbar\coth(\hbar\omega_m/2k_BT)/2\omega_m$ the quantum-thermal variance of the mass-weighted mode coordinate. Configuration k is

$$u_i^{(k)} = \frac{1}{\sqrt{M_i}} \sum_m s_{km}\,\sigma_m\,e_{im}, \qquad s_{km}\in\{-1,+1\}, \tag{7}$$

so that every configuration reproduces every mode variance exactly; the off-diagonal products $s_{km}s_{km'}\sigma_m\sigma_{m'}$ vanish only on average over sign patterns (Supplementary Material, Sec. S10). The signs are the only design variables. Equations (5)–(7) define the designs introduced in this work. Writing $\mathbf{X}$ for the force design matrix of the space-group adapted FC2 basis (Sec. III A), the signs of configuration k are chosen by greedy single-flip ascent of $\log\det(\mathbf{X}^{\rm T}\mathbf{X})$ given configurations $1,\dots,k{-}1$ (sequential D-optimal design[18]), so that every prefix of the sequence is itself a design. Each configuration is evaluated together with its inverse $-\mathbf{u}^{(k)}$ (antithetic pairing[19]). Because the third-order force is even in $\mathbf{u}$, the pair cancels it exactly from the linear FC2 fit, whereas the quartic renormalization that defines the effective FC2 is retained. A budget of n force calls thus corresponds to $n/2$ designed config-

urations. The trial model enters only through ω_m, $\mathbf{e}_m$, and σ_m, as it does for canonical sampling.

E. Bias and variance of the SOTC-D estimator

Writing the potential in normal coordinates as $V = \frac{1}{2}\sum \omega_a^2 Q_a^2 + \frac{1}{6}\sum V_{abc} Q_a Q_b Q_c + \frac{1}{24}\sum V_{abcd} Q_a Q_b Q_c Q_d$, the canonical target of Eq. (2) is $\Phi_{ae} = \omega_a^2 \delta_{ae} + \frac{1}{2}\sum_b V_{abbe}\sigma_b^2$ to fourth order in the displacements. Averaging the regression coefficient of Eq. (7) over independent sign patterns, the only moment that differs from the Gaussian is $\langle Q_e^4\rangle = \sigma_e^4$ instead of $3\sigma_e^4$, so that

$$\Phi_{ae}^{\text{SOTC-D}} - \Phi_{ae} = -\tfrac{1}{3} V_{aeee}\sigma_e^2. \tag{8}$$

This is one diagonal term of the renormalization sum over the M commensurate modes, so the bias is $O(1/M)$ relative to the quartic renormalization and vanishes for large cells. D-optimal signs are not independent, so Eq. (8) describes the design on average; the residual is part of the variance discussed below. The bias is also removed exactly in finite cells if the squared amplitudes of each mode are varied between configurations with the Gaussian moments $\langle Q_e^2\rangle = \sigma_e^2$ and $\langle Q_e^4\rangle = 3\sigma_e^4$ (moment-matched SOTC-D, Sec. V H), at the price of a larger variance at small budgets. Eq. (8) is derived here (Supplementary Material, Sec. S4) and was checked by exact enumeration over sign patterns; the averaged-Hessian identity follows from Stein's lemma[16], and its equivalence with the SCHA stationarity condition is standard[6].

Under $\mathbf{u} \to -\mathbf{u}$ the cubic force is even and the harmonic and quartic forces are odd. For a $\pm$ pair the cubic force therefore lies in the null space of the linear fit and cancels exactly, for any FC3; for unpaired canonical samples it enters as noise of order $V_{abc}\sigma^2$. The remaining variance of the least-squares estimator is $\propto (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}$, and maximizing $\det \mathbf{X}^{\mathrm{T}}\mathbf{X}$ minimizes the volume of its confidence ellipsoid[18]. Fixed amplitudes remove the amplitude variance of $\mathbf{X}^{\mathrm{T}}\mathbf{X}$, which for canonical draws makes weakly excited modes poorly determined at small budgets.

F. Self-consistent design (SOTC-D-SC)

The trial enters SOTC-D only through ω_m, $\mathbf{e}_m$, and σ_m, and can be updated during the design. After each $\pm$ pair, all data so far are refitted; while the data do not yet determine every coefficient, the update is the minimum-norm correction of the trial coefficients consistent with the data. The modes and amplitudes of the next pair are computed from the updated model, and its signs maximize $\det(\mathbf{X}_{\text{prev}}^{\mathrm{T}}\mathbf{X}_{\text{prev}} + \mathbf{X}^{\mathrm{T}}\mathbf{X})$ given all previous configurations. This converges to the self-consistent target without discarding the forces of earlier iterations, unlike sTDEP, which redraws each iteration. It also repairs a defective trial: modes that are unstable in the trial are omitted from the first pair (as in canonical sampling) but acquire the correct amplitude once the fitted model stabilizes them. In the DFT benchmark below we use the simplest version, *bootstrapped* SOTC-D, with a single update: the first pair is designed from the harmonic trial, and the remaining pairs are designed from the model refitted to that pair, with signs chosen sequentially given all earlier pairs. The name SOTC-D-SC is reserved for the version that updates the trial after every pair, which is used on the model potentials of Sec. V H.

III. IMPLEMENTATION

A. Effective FC2 regression

Forces are evaluated for the undisplaced reference and for each designed configuration in the same cell, and the response $\Delta\mathbf{F} = \mathbf{F}(\mathbf{u}) - \mathbf{F}(\mathbf{0})$ is fitted to the model below. At the ideal positions used here $\mathbf{F}(\mathbf{0})$ vanishes by site symmetry; the subtraction removes residual numerical forces, while the effect of using a lattice parameter that is not the PBE minimum is assessed separately (Sec. IV). The model is

$$\Delta F_i^{\alpha} = -\sum_{j\neq i,\beta} \Phi_{ij}^{\alpha\beta}\left(u_j^{\beta} - u_i^{\beta}\right), \tag{9}$$

with one 3×3 tensor per translationally distinct pair vector within the cutoff r_{cut} and Newton's third law $\mathbf{\Phi}(-\mathbf{R}) = \mathbf{\Phi}(\mathbf{R})^{\mathrm{T}}$ built into the parametrization. The tensors are obtained by ridge regression

$$\hat{\mathbf{p}} = -\left(\mathbf{G}^{\mathrm{T}}\mathbf{G} + \lambda\mathbf{I}\right)^{-1}\mathbf{G}^{\mathrm{T}}\Delta\mathbf{F}, \tag{10}$$

with $\lambda = 10^{-3}$ Å^2. Equation (9) is the standard pair-force model of TDEP-type fits[8] and Eq. (10) standard ridge regression. Two physical priors are then applied. First, each pair tensor is projected onto the axially symmetric form $\mathbf{\Phi} = \alpha\hat{\mathbf{R}}\hat{\mathbf{R}} + \beta(\mathbf{I} - \hat{\mathbf{R}}\hat{\mathbf{R}})$ and α, β are averaged over each shell (the "axial projection"). Second, the on-site blocks are set by the acoustic sum rule, $\mathbf{\Phi}_{ii} = -\sum_{j\neq i}\mathbf{\Phi}_{ij}$. The axial form is exact for bonds along threefold or fourfold axes (bcc first neighbors, fcc and rocksalt second neighbors, rocksalt first neighbors) but omits the third independent constant allowed by symmetry for fcc first-neighbor $\langle 110\rangle$ bonds. Because these priors can bias the result, Sec. V A reports the same data refitted without the projection, with different λ and cutoffs, and in a complete space-group, permutation, and sum-rule adapted FC2 basis constructed with symfc[20], solved by unregularized least squares; this fit is used for SOTC-D and for all matched canonical baselines. For that basis, the rank and condition number of the column-normalized design matrix are reported.

The symfc basis is adapted to the space group of the *supercell*, which for the low-symmetry SOTC cells is a subgroup of the crystal space group and admits 30–90

parameters. We therefore also use the FC2 basis of the full crystal space group, constructed with hiPhive[21] from the primitive cell and mapped to the supercell, with a pair cutoff limited to half the shortest supercell image distance so that no pair interaction is aliased. It has 3 (fcc), 7 (bcc), 8 (rocksalt) parameters at the same physical range (Sec. V E). For the same data, fewer parameters at the same physical range reduce the variance of every estimator.

B. SOTC-C workflow and phonon properties

Iteration n designs configurations for targets computed from the current phonon model (a Debye model at $n = 0$), evaluates forces, fits the FC2 model, and recomputes $\bar{C}_2$ from Eq. (4) on an $8 \times 8 \times 8$ q mesh. The targets are linearly mixed with weight 0.5; an update from a spectrum with more than 1% unstable modes is rejected. The loop stops when the root-mean-square change in $\bar{C}_2$ falls below 3×10^{-3} Å^2 after a material-specific minimum number of iterations, and the most stable iteration is retained. Each iteration recomputes the reference, so the number of force calculations is $N_{\mathrm{iter}}(N_{\mathrm{ens}} + 1)$.

Dynamical matrices are Fourier interpolated from the real-space FC2. Isochoric heat capacity C_V, vibrational entropy S_{vib}, free energy F_{vib}, and zero-point energy follow from the harmonic partition function of the fitted frequencies, omitting modes below 0.05 THz (including imaginary ones); the spectral Debye temperature is the standard second-moment value $T_D = (\hbar/k_B)\sqrt{5\langle\omega^2\rangle/3}$. These are quantities of the effective harmonic model at the cell volume used. The electronic heat capacity is not included. Thermal expansion, with the anharmonic correction to the free energy, is computed for Al only (Sec. V K).

IV. COMPUTATIONAL DETAILS

A. Electronic structure

Forces were computed with Quantum ESPRESSO (QE) 7.5[22], the PBE functional[23], and pseudopotentials of the standard solid-state pseudopotential (SSSP) efficiency library[24] (optimized norm-conserving Vanderbilt (ONCV) for Ag, Au, Mo; projector augmented-wave (PAW) for Al, Cu, Mg, O, Pb; ultrasoft for Na, Cl, Te, W). Wave-function cutoffs were 40 (Al), 50 (Ag), 45 (Au, Cu), 40 (Mo), 30 (W), 60 (NaCl), 80 (MgO), and 44 Ry (PbTe); charge-density cutoffs were four (ONCV) or eight (PAW, ultrasoft) times larger. Metals used 0.02 Ry Marzari–Vanderbilt smearing[25]; the rocksalt compounds used fixed occupations. The SCF threshold was 10^{-8} Ry for the primary SOTC runs and their same-cell controls and 10^{-9} Ry for all other references.

TABLE I. Correlator-matched SOTC runs (SOTC-C). a: lattice parameter; n: supercell atoms; N_{iter}, N_{ens}: iterations and configurations per iteration; $N_{\mathrm{force}} = N_{\mathrm{iter}}(N_{\mathrm{ens}}+1)$ QE calls; p: parameters of the space-group adapted FC2 basis within r_{cut} in the SOTC cell.

	a (Å)	T_{des} (K)	n	r_{cut} (Å)	N_{iter}	N_{ens}	N_{force}	p
Al	4.05	300	32	4.5	2	5	12	30
Ag	4.12	1000	32	4.5	4	5	24	30
Au	4.13	1200	32	4.5	2	1	4	30
Cu	3.67	1200	32	4.5	4	8	36	90
Mo	3.16	1500	64	5.0	2	1	4	78
W	3.18	1500	64	5.0	4	3	16	78
NaCl	5.73	700	64	4.3	5	3	20	53
MgO	4.26	1500	64	3.2	4	3	16	53
PbTe	6.51	700	72	5.0	4	5	24	53

B. Correlator-matched SOTC runs

The lattice parameters (Table I) are experimental values near the design temperature T_{des}; ideal positions were used. The supercells were chosen by enumerating Hermite normal forms at the target size: 32 atoms for the fcc metals, 64 for the bcc metals and NaCl and MgO, and 72 for PbTe. Metals used a $2 \times 2 \times 2$ supercell k mesh, rocksalt compounds Γ only. Table I lists the cutoffs, iteration counts, and force-call accounting. The SOTC-C model of each crystal is the fit of Eq. (9) to the configurations of the iteration selected by the workflow, i.e., the most stable converged one (Sec. III B); all SOTC-C results below are refits of the stored force–displacement data.

C. Reference and control calculations

Three classes of additional calculations were performed. *Converged harmonic references* used central finite displacements (±0.01 Å) with the open-source finite-displacement phonon code phonopy[26] in $5\times5\times5$ primitive supercells (125 atoms) for the metals and $4 \times 4 \times 4$ (128 atoms) for the rocksalt compounds, with $3 \times 3 \times 3$ and $2\times2\times2$ k meshes, respectively, at the production lattice parameters. The same calculations in $3 \times 3 \times 3$ primitive supercells (27 or 54 atoms) were also performed, together with references at the PBE equilibrium lattice parameter obtained from third-order Birch–Murnaghan fits to primitive-cell energies at seven lattice parameters within ±3%, and at the experimental room-temperature lattice parameter for Al, Cu, and Au. The 128-atom PbTe states used parallel-orbital diagonalization with local Thomas–Fermi mixing, because Davidson diagonalization failed and plain mixing did not converge in this cell. *Same-cell controls* used exactly the primary SOTC supercell and electronic settings: a central FD harmonic FC2, and 48 quantum-canonical configurations (24 $\pm\mathbf{u}$ pairs) drawn at T_{des} from that harmonic model. Pairs

1–20 form a 40-snapshot training pool and pairs 21–24 a held-out test set. The *effective reference* is the symfc FC2 fitted to all 40 training snapshots; *matched canonical baselines* are the space-group adapted least-squares fit (Sec. III A) applied to 20 random subsets of the pool with exactly as many configurations as the SOTC-C fit being compared, or of complete ± pairs. Because a canonical subset shares configurations with a reference fitted to the whole pool, dispersion errors in the matched comparison are measured against an effective FC2 fitted to the $40 - n$ pool snapshots *disjoint* from the scored design (the complement of a canonical subset, or a random $(40-n)$-subset for SOTC-D), and held-out forces are used as a reference-free measure. Finally, *SOTC-D designs* of eight configurations (16 force calls with their inverses) were generated from the same-cell harmonic FD model, i.e., from exactly the trial model used for the canonical baselines, and evaluated in the same cell and settings.

Two further competitors were generated from the same trial and evaluated in the same way (16 configurations each): *special displacements* (ZG)[15], in which the real, mass-normalized commensurate modes of the half Brillouin zone are excited at the canonical amplitude with the orthogonal Walsh sign patterns of Ref. 15, and a *thermal rattle* as used for hiPhive training sets[21], in which every atom is displaced independently with the canonical per-atom variance. To test the trial update (Sec. II F), the effective FC2 of the 40 iteration-0 snapshots was used as a new trial (*iteration 1*, one sTDEP step), from which 48 further canonical configurations (40 training, 8 test) and 16 SOTC-D configurations were computed; and a bootstrapped SOTC-D set was computed whose first pair is that of the iteration-0 SOTC-D and whose seven further pairs are designed from the model refitted to the first pair (Sec. II F). For a DFT-level sTDEP comparison, two-iteration sTDEP runs were computed with our implementation of the protocol of Ref. 12, using the regression of Sec. III A rather than the TDEP code, for budgets of 8 and 16 calls (three seeds each): half of the budget as canonical ± pairs from the harmonic trial (taken from the iteration-0 pool), and half as new canonical ± pairs drawn from the model refitted to them (36 additional force calls per crystal). For Al, SSCHA and SOTC-D-SC (Sec. II F) were also run with DFT forces in the same 32-atom cell. SSCHA used python-sscha[6] with the whole cell as the unit cell (Γ point only), the same harmonic trial, and the minimizer settings of Sec. V H, with populations of $N = 16$, 32, 64, and 256 configurations; SOTC-D-SC was run with three seeds up to 32 force calls. For comparison, the SSCHA force constants are projected onto the same supercell basis. As an AIMD-TDEP baseline for Al, four Born–Oppenheimer molecular-dynamics trajectories were run with QE in the same cell and with the same electronic settings (Verlet integration, 3.9 fs time step, stochastic velocity rescaling at 300 K with a 77 fs time constant), each started from an independent canonical configuration and run for 1307–1372 steps. After 250 equilibration steps, the FC2 was fitted with the same basis and regression to the first n production steps of each trajectory; every MD step is counted as one force call. Iteration-1 results are scored against the iteration-1 effective FC2. For the thermal expansion of Al, the 32-atom cell was scaled to seven lattice parameters from 4.000 to 4.150 Å in steps of 0.025 Å. At each, the harmonic FD trial (four displacements and the undisplaced cell) was computed, and SOTC-D-SC was run with 16 force calls at 100, 300, 400, 500, 700, and 900 K, twice: once with design seeds that differ between volumes and once with the same seeds at all volumes. The harmonic FD FC2 was also computed on the same volumes in the 125-atom cell with the large-cell settings above.

The same protocol was applied to two further classes of crystal: covalent diamond Si (600 K, 64-atom conventional $2 \times 2 \times 2$ cell, $r_{\mathrm{cut}} = 5.0$ Å, ONCV pseudopotential[24], 30/120 Ry, $2 \times 2 \times 2$ k mesh) and hexagonal close-packed Mg (600 K, $4 \times 4 \times 2$ 64-atom cell, $r_{\mathrm{cut}} = 4.6$ Å, PAW, 40/320 Ry, $3 \times 3 \times 3$ k mesh), with converged references in 128-atom ($4 \times 4 \times 4$ primitive) and 150-atom ($5 \times 5 \times 3$) cells. For these crystals no SOTC-C runs were made, and only the same-cell protocol above is used.

Non-analytic corrections for NaCl, MgO, and PbTe used Born effective charges and electronic dielectric tensors from QE density-functional perturbation theory (DFPT) in the primitive cell at the production lattice parameters (60/480 Ry, $8 \times 8 \times 8$ k mesh), applied through the mixed-space formulation in phonopy[26]. Phonon densities of states and thermodynamic quantities were evaluated on $12 \times 12 \times 12$ (comparisons) or $30 \times 30 \times 30$ (DOS) meshes; U_{iso} was evaluated on a shifted $24 \times 24 \times 24$ mesh, which changes the converged-reference value for Al by less than 1% relative to $60 \times 60 \times 60$.

V. RESULTS

A. Identifiability and priors of SOTC-C fits

Table II summarizes the identifiability of the nine SOTC-C data sets. In the space-group adapted basis, the force design of every single iteration has full column rank, with 30 (fcc metals), 53 (rocksalt), 78 (bcc), or 90 (Cu) FC2 parameters and normalized condition numbers of 2.9–17.1. This is possible because a single force calculation in an n-atom cell gives $3n$ force components (96–216 here), each a linear equation in the FC2 parameters, and because a collective displacement of all atoms makes many of them linearly independent; the symmetry-adapted basis reduces the unknowns to 30–90. A few collective SOTC configurations therefore identify the FC2 model of these cells completely once the space-group symmetry of the supercell and the acoustic sum rule are imposed, without any further prior. The parameter count is set by the number of neighbor shells within r_{cut}, not by the number of configurations. For Cu ($a = 3.67$ Å) the 4.5 Å cutoff includes the third fcc

TABLE II. Identifiability and priors of the SOTC-C force data. Rank/p and the largest condition number over all iterations refer to the unregularized space-group adapted FC2 design of each single iteration. f_{unst} is the fraction of unstable 12^3-mesh modes, with the axial prior and without the axial projection, in the worst iteration. Δ_λ and Δ_r are the ranges (THz) of the mesh RMSE against the converged FD reference over $\lambda \in [0, 0.1]$ and $r_{\mathrm{cut}} \pm 0.5$ Å for the pooled data.

	rank/p	cond.	f_{unst} axial	f_{unst} no proj.	Δ_λ, Δ_r
Al	30/30	5.2	0.000	0.010	0.04, 0.10
Ag	30/30	5.7	0.000	0.063	0.00, 0.02
Au	30/30	10.7	0.159	0.334	0.09, 0.04
Cu	90/90	15.3	0.000	0.069	0.00, 0.09
Mo	78/78	17.1	0.000	0.075	0.04, 0.05
W	78/78	4.5	0.000	0.002	0.01, 0.06
NaCl	53/53	3.9	0.000	0.106	0.00, 0.18
MgO	53/53	2.9	0.000	0.033	0.01, 0.83
PbTe	53/53	3.1	0.000	0.080	0.00, 0.03

shell at $a\sqrt{3/2} = 4.49$ Å, which for Al, Ag, and Au lies beyond it (4.96–5.06 Å); the Cu model therefore has 90 rather than 30 parameters in the same 32-atom cell; because this shell lies only 0.005 Å inside the cutoff, the Cu parameter count is sensitive to the choice of r_{cut}. The SOTC-C runs also differ in the number of configurations per iteration, $N_{\mathrm{ens}} = 1$ (Au, Mo), 3 (W, NaCl, MgO), 5 (Al, Ag, PbTe), or 8 (Cu), as listed in Table I; with $N_{\mathrm{ens}} = 8$ each Cu iteration alone has full rank 90.

The regression prior matters more than identifiability. Without the axial projection, the ridge fits are unstable in every material (up to 33% unstable modes for Au), because the general 3×3 pair tensors of Eq. (9) are not symmetry constrained. With the projection, eight of the nine SOTC-C models are stable. The exception is Au, whose two iterations each used a single configuration: 0.7% and 15.9% of the mesh modes are unstable, and the selected first iteration therefore has 0.7% unstable modes (minimum −0.95 THz; Supplementary Table S5). Pooling the two Au configurations removes it (minimum frequency 0.27 THz), and the converged harmonic FD reference at the same lattice parameter is stable (minimum 0.36 THz), so the instability is an artifact of the one-configuration fit rather than a property of PBE Au. The ridge parameter changes the results by at most 0.09 THz between $\lambda = 0$ and 0.1 Å^2; the cutoff bracket matters more, particularly for MgO, where lowering r_{cut} by 0.5 Å to 2.7 Å removes the second-neighbor shell at 3.0 Å and changes the RMSE by 0.83 THz (Table II).

B. Reference accuracy: cell size and temperature

Table III separates the error sources (overview in Supplementary Fig. S2). Against the converged large-cell harmonic references, the harmonic FD FC2 in the 27–54-atom ($3 \times 3 \times 3$) cells deviates by 0.05–0.48 THz, and that in the SOTC cells by 0.06–0.49 THz. Supercell truncation is therefore significant for Al (0.49 THz in the 32-atom cell) and MgO (0.49 THz), and for PbTe, whose 54- and 72-atom FD models have 1.9% unstable modes, whereas the 128-atom reference is stable (lowest frequency 0.46 THz). The converged Al reference gives 5.72 and 9.35 THz for the TA and LA modes at X and 4.25 and 9.66 THz at L. The Born–von Kármán model fitted to the inelastic neutron scattering spectrum of Al at 10 K (eight shells)[27,28] gives 5.75 and 9.45 THz at X and 4.21 and 9.72 THz at L; the converged reference agrees with all four within 1.1% (see also the dispersion of Ref. 29). The 300 K model of the same work, which includes thermal expansion, gives 5.39, 9.37, 4.16, and 9.43 THz. Its transverse branches are softer than both the converged reference and the 300 K effective FC2 of the 32-atom cell (TA(X) 5.39 against 5.72 and 6.01–6.29 THz, the degeneracy being lifted by the cell). Most of the stiffening of the effective FC2 is already present in the harmonic FD model of the same cell (5.87–6.09 THz) and is therefore a truncation effect (Supplementary Fig. S4). By contrast, the $3\times3\times3$ FD model places TA(L) at 4.91 THz, 15% higher: in small cells the truncation error is of the same size as the thermal effects studied here. The effective FC2 at T_{des} differs from the harmonic one by 0.16 (W, 1500 K) to 1.36 THz (MgO, 1500 K) (last column of Table III); the effective spectra are stiffer, with spectral Debye temperatures higher by 7 K (PbTe) to 54 K (MgO) for all materials except Al, Mo, and W, where the two agree within 4 K. Comparisons of high-temperature effective models with harmonic references therefore conflate this physical renormalization with fitting error. The effective references themselves are converged to 0.02–0.20 THz between 20 and 40 snapshots.

Measured against the same-cell effective FC2, the SOTC-C models deviate by 0.12–0.77 THz (0.12–0.44 THz excluding Au), and pooling all SOTC-C iterations improves Au to 0.32 THz. For these nine crystals SOTC-D with eight force calls deviates by 0.04–0.31 THz, a 1.3- to 9.8-fold improvement over SOTC-C (1.3-fold for NaCl, 0.14 versus 0.18 THz) for all materials except Cu. For Cu the SOTC-C model, fitted to the eight configurations of the selected iteration of a 36-call workflow, is more accurate (0.24 versus 0.31 THz).

C. Matched comparison with canonical sampling and other designs

Figure 1 and Table IV compare SOTC-D with canonical sampling with least-squares fitting (as used in sTDEP), special displacements, and thermal rattle at equal numbers of force calls, with identical cells, electronic settings, trial model, and fit. The comparison isolates the effect of how the configurations are chosen. The advantage of SOTC-D is largest at the smallest budgets: with two force calls, a single designed ± pair excites every commensurate mode at its thermal amplitude

TABLE III. Phonon-mesh RMSE (THz, 12^3 mesh) of FC2 models for the nine benchmarks. Columns 2–4 are measured against the converged large-cell harmonic FD reference; columns 5–8 against the effective FC2 fitted to 40 canonical snapshots at T_{des} in the SOTC cell. "SOTC-C" is the correlator-matched model (selected iteration, axial prior); "SOTC-D" is the D-optimal antithetic design with the space-group adapted fit at 8 force calls. The last column is the RMSE between the harmonic and effective models (finite-temperature renormalization). Because the reference here is the full 40-snapshot fit, SOTC-D values differ by up to 0.02 THz from those of Table IV, whose reference is disjoint from the scored design.

	vs converged harmonic FD			vs same-cell effective FC2			
	FD 3^3	FD SOTC cell	SOTC-C	SOTC-C	SOTC-C pooled	SOTC-D (8)	harmonic–effective
Al	0.30	0.49	0.28	0.30	0.24	0.08	0.24
Ag	0.05	0.06	0.71	0.29	0.25	0.09	0.46
Au	0.09	0.12	0.91	0.77	0.32	0.12	0.58
Cu	0.07	0.07	0.73	0.24	0.29	0.31	0.67
Mo	0.38	0.27	0.44	0.44	0.11	0.04	0.28
W	0.20	0.15	0.23	0.12	0.08	0.04	0.16
NaCl	0.16	0.18	0.68	0.18	0.16	0.14	0.71
MgO	0.48	0.49	1.38	0.37	0.43	0.28	1.36
PbTe	0.22	0.24	0.47	0.25	0.30	0.10	0.32
Si	–	0.07	–	–	–	0.04	0.26
Mg	–	0.05	–	–	–	0.03	0.28

and is free of cubic-force noise, whereas two canonical draws excite some modes weakly and carry the full cubic noise. Accordingly, with two calls SOTC-D gives the lowest dispersion and held-out force errors of all five designs for every crystal except Cu in dispersion, whose 90-parameter model no two-call design determines, and PbTe in force, whose trial is defective (see below). The ranking in dispersion holds in 8 of 11 crystals at four calls (Supplementary Table S7): paired canonical sampling is better for NaCl and PbTe (0.24 versus 0.25 and 0.13 versus 0.16 THz), and special displacements are better for MgO (0.395 versus 0.405 THz). At eight calls it holds in 10 of 11 crystals, where the Si and Mg results are ties within 0.005 THz; in held-out force SOTC-D is best in 8 of 11 crystals at eight calls and within 2.1% of the best design in the others. At 16 calls SOTC-D is best in only 4 of 11 crystals: the sampling error of every design approaches the noise of the disjoint reference, and the designs can no longer be distinguished.

The special-displacement design, which also fixes the mode amplitudes but uses orthogonal rather than information-optimal signs, is worse than SOTC-D in all 11 crystals at eight calls, by up to a factor of six for Si (0.29 versus 0.05 THz); with two calls in Cu it does not identify the model. Its Walsh patterns assign equal signs to modes that are adjacent in the frequency ordering and are not chosen with respect to the force-constant basis, so that they leave the information matrix less well conditioned than the D-optimal signs. Thermal rattle, a common training set of hiPhive workflows, is worse than SOTC-D in every crystal and the worst of all designs in 9 of 11 (0.06–0.89 THz at eight calls). Independent atomic displacements have the canonical on-site variance but no interatomic correlations; the fitted force constants are then the averaged Hessian of an uncorrelated ensemble, which differs from the canonical target of Eq. (2). The held-out forces of NaCl, in contrast, are insensitive to the design: the canonical, SOTC-D, and rattle designs lie within 2% of each other at eight calls.

The one material in which the iteration-0 SOTC-D is not the best design at eight calls is PbTe (0.11 versus 0.10 THz). The cause is the trial: the same-cell harmonic FD model of PbTe has two unstable commensurate modes (-1.72 and -0.49 THz), which are omitted from SOTC-D and from canonical draws alike, so that every iteration-0 design underexcites the soft optical branch. This is a deficiency of the trial, not of the design, and is removed by the trial update of the next subsection. The fixed amplitudes also make the small-budget fits more robust: every SOTC-D model of Au with four or more force calls is stable, and with two calls 0.2% of the mesh modes are unstable, whereas canonical sets of two calls produce unstable modes in most draws.

D. Trial update and equal-cost comparison

Updating the trial from the harmonic FC2 to the iteration-1 effective FC2 changes the target by 0.03–0.28 THz and, more importantly, changes the amplitudes at which the modes are excited. Table V and Fig. 2 repeat the comparison with this updated trial. With a trial close to the effective FC2, the advantage of the design becomes uniform: SOTC-D has the lower dispersion error at four and eight calls in every crystal, and at two calls in all except Cu, for the reason given above; its held-out force error is the lower one in 11, 10, and 10 of the 11 crystals at two, four, and eight calls (Table V and Supplementary Table S9). PbTe changes from the only crystal without a gain to the one with the largest gain (0.08 versus 0.16 THz with two calls, 0.04 versus 0.08 THz with eight). The updated trial is dynamically stable, so the soft transverse optical branch, which the harmonic trial could not excite, is now sampled at its thermal ampli-

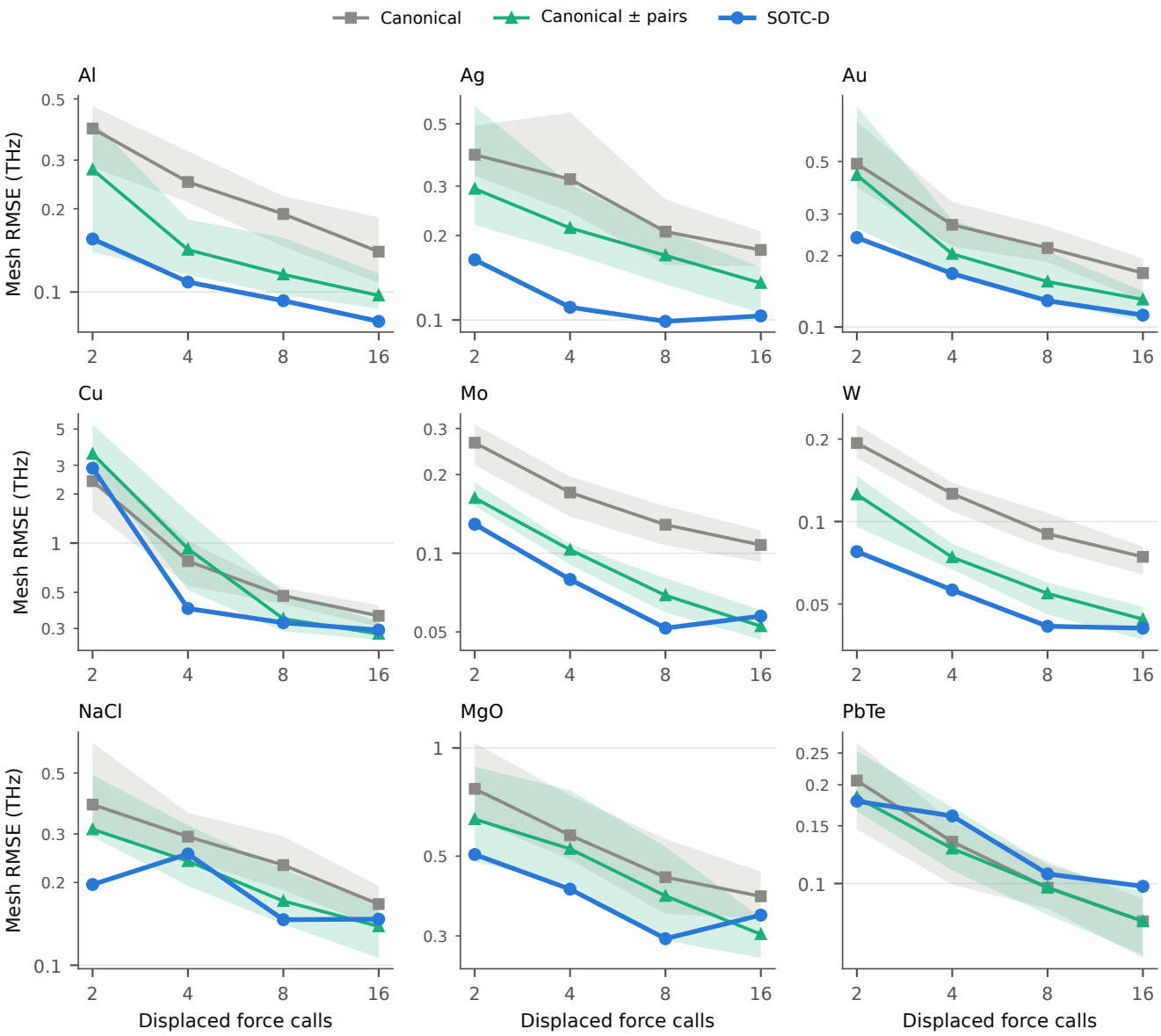


FIG. 1. Phonon-mesh RMSE against a disjoint same-cell effective FC2 versus the number of displaced force calls, for SOTC-D (blue circles), random canonical subsets (gray squares), and random canonical $\pm\mathbf{u}$ pairs (green triangles), all designed from the same-cell harmonic FD trial (iteration 0). Lines are medians and bands the 10–90% range over 20 random draws. All models use the space-group adapted fit in the SOTC cell with identical electronic settings. Shown are the nine cubic crystals; Si and Mg are included in Table IV.

tude.

Because the iteration-1 trial itself requires 40 force calls, a comparison at equal total cost must start every protocol from the harmonic trial. Bootstrapped SOTC-D (Sec. II F) requires no force calls beyond the budget: its first pair is the iteration-0 SOTC-D pair, and the model refitted to this pair defines the modes and amplitudes of the remaining pairs. Measured against the iteration-1 effective FC2, it is more accurate than one-shot canonical pairs from the harmonic trial (a single sTDEP iteration) in all 11 crystals at 4, 8, and 16 calls, and at two calls in the ten crystals whose model a single pair identifies (not Cu). At eight calls its error is 0.02–0.31 THz against 0.04–0.39 THz, and the largest gains are for the crystals with the largest trial update: Au (0.07 versus 0.25 THz), MgO (0.19 versus 0.36), NaCl (0.10 versus 0.18), and PbTe (0.06 versus 0.12). It is also better than one-shot SOTC-D for the same crystals, so that the update, and not only the design, is responsible; for Cu, Mo, and W, whose trial changes least relative to the cell error, one-shot SOTC-D is slightly more accurate at eight calls (0.28, 0.044, and 0.041 versus 0.31, 0.055, and

TABLE IV. Matched comparison in the SOTC cell at equal numbers of displaced force calls, all designed from the same-cell harmonic FD trial and fitted in the supercell space-group basis. D: SOTC-D; C: random canonical subsets; P: random canonical $\pm\mathbf{u}$ pairs (medians over 20 draws); Z: special displacements with Walsh sign patterns (ZG); R: thermal rattle (independent Gaussian atoms with the canonical per-atom variance). Upper block: held-out force RMSE on 8 canonical test snapshots relative to that of the 40-snapshot effective FC2. Lower block: mesh RMSE (THz) against an effective FC2 fitted to $40 - n$ canonical snapshots disjoint from the scored design. The best entry is bold; the last row counts the materials in which D is best. Entries >10 THz (>100 in force) are singular fits of designs that do not identify the model.

	2 calls					8 calls				
	D	C	P	Z	R	D	C	P	Z	R
Held-out force RMSE / floor										
Al	**1.016**	1.100	1.047	1.064	1.087	**0.999**	1.017	1.010	1.017	1.032
Ag	**1.008**	1.103	1.071	1.123	1.444	**0.999**	1.028	1.016	1.011	1.042
Au	**1.007**	1.099	1.094	1.124	1.302	**1.009**	1.026	1.013	1.024	1.083
Cu	**1.614**	1.739	2.173	>100	3.044	**1.002**	1.071	1.033	1.045	1.079
Mo	**1.022**	1.152	1.069	1.250	1.120	**0.999**	1.028	1.008	1.012	1.021
W	**1.018**	1.151	1.054	1.129	1.103	**1.001**	1.024	1.007	1.010	1.008
NaCl	**1.005**	1.010	1.009	1.057	1.014	1.005	1.001	1.001	0.997	**0.985**
MgO	**1.021**	1.085	1.059	1.054	1.355	1.006	1.015	1.012	**0.999**	1.077
PbTe	1.038	1.054	1.060	1.082	**1.028**	1.016	1.003	**1.002**	1.021	1.147
Si	**0.999**	1.015	1.006	1.024	1.133	**0.998**	1.002	1.000	1.036	1.151
Mg	**1.002**	1.016	1.005	1.006	1.033	**1.000**	1.003	1.001	1.002	1.021
D best			10/11					8/11		
Mesh RMSE vs disjoint effective FC2 (THz)										
Al	**0.16**	0.39	0.28	0.30	0.34	**0.09**	0.19	0.12	0.13	0.22
Ag	**0.16**	0.39	0.29	0.47	1.26	**0.10**	0.21	0.17	0.20	0.27
Au	**0.24**	0.49	0.44	0.59	0.65	**0.13**	0.22	0.16	0.21	0.35
Cu	2.88	**2.40**	3.52	>10	4.43	**0.33**	0.48	0.34	0.54	0.64
Mo	**0.13**	0.26	0.16	0.36	0.20	**0.05**	0.13	0.07	0.08	0.12
W	**0.08**	0.19	0.13	0.23	0.15	**0.04**	0.09	0.05	0.05	0.06
NaCl	**0.20**	0.38	0.31	1.58	0.97	**0.15**	0.23	0.17	0.32	0.89
MgO	**0.51**	0.77	0.63	0.83	1.95	**0.29**	0.44	0.39	0.34	0.81
PbTe	**0.18**	0.21	0.18	0.25	0.23	0.11	**0.10**	0.10	0.12	0.36
Si	**0.07**	0.16	0.10	0.23	0.55	**0.05**	0.09	0.05	0.29	0.64
Mg	**0.07**	0.14	0.08	0.09	0.23	**0.04**	0.07	0.04	0.05	0.20
D best			10/11					10/11		

0.045 THz). Starting from the harmonic trial alone, eight bootstrapped SOTC-D calls are more accurate than sixteen one-shot canonical calls for 9 of the 11 crystals (all except Cu and W).

The iterated canonical protocol, sTDEP[12], was run with DFT forces in all eleven crystals at the same budgets (Table VI), in our implementation with the regression of Sec. III A. With eight calls, bootstrapped SOTC-D has the lowest error in all 11 crystals, with errors lower by up to a factor of 2.3 than sTDEP fitted to its last iteration and up to 3.8 (Au) than sTDEP fitted to both iterations (pooled); with 16 calls it is best in 9 of 11, the exceptions being Cu (0.25 versus 0.18 THz for pooled sTDEP) and Mo (0.055 versus 0.048). Eight bootstrapped SOTC-D calls are more accurate than sixteen sTDEP calls with either fit in 7 of 11 crystals (not Cu, Mo, W, Si). In the crystal-symmetry basis, where all errors are smaller, the margins shrink: bootstrapped SOTC-D is best in 8 of 11 crystals at eight calls and in 6 of 11 at sixteen, and pooled sTDEP is better for Cu at both budgets (Supplementary Table S3). The advantage of the design is thus largest at the smallest budgets and for the crystals whose harmonic trial is furthest from the effective FC2 (Au, NaCl, MgO, PbTe). Both iterated protocols aim at the self-consistent FC2, of which the iteration-1 FC2 used for scoring is only the first approximation; the model potentials of Sec. V H, for which the self-consistent target is known exactly, remove this limitation.

For Al, SSCHA and SOTC-D-SC were also compared with DFT forces. Measured against the iteration-1 effective FC2, which lies 0.27 THz from the harmonic trial, SOTC-D-SC reaches median errors of 0.097, 0.075, 0.055, and 0.050 THz with 4, 8, 16, and 32 force calls (three seeds). SSCHA stops after its first population for every population size, because the statistical error of its free-energy gradient exceeds the gradient itself; its result approaches the effective FC2 only as the population grows, with errors of 0.217, 0.193, 0.131, and 0.092 THz for populations of 16, 32, 64, and 256 force calls. Against this reference, SOTC-D-SC with four force calls is as close to the effective FC2 as SSCHA with 256. The test is limited by the reference, however. Measured instead against the 256-configuration SSCHA result (projected onto the same basis, which changes it by 0.23 THz; it is 0.24 THz

from the harmonic trial), SSCHA with 64 configurations and the iteration-1 FC2 lie at 0.09 THz, SOTC-D-SC at 0.12, 0.12, 0.11, and 0.10 THz with 4, 8, 16, and 32 calls, and the smaller SSCHA populations, one-shot canonical pairs, and two-iteration sTDEP at 0.12–0.19 THz. With DFT forces in this cell, differences below about 0.1 THz between these protocols are therefore not resolved.

AIMD-TDEP converges more slowly in force calls and to a different limit. Measured against the fit to all 4370 production steps, the median error of a single trajectory is 0.27 THz after 282 steps (including the 250 equilibration steps), 0.12 THz after 378, 0.097 THz after 506, and 0.056 THz after 1274 steps. SOTC-D-SC reaches a comparable error relative to its own target with four force calls, although that target, the 40-snapshot iteration-1 FC2, is itself uncertain at the 0.1 THz level (see above). The converged AIMD-TDEP FC2 lies 0.28 THz from the iteration-1 effective FC2 and 0.27 THz from the harmonic trial. The mean mesh frequency is 5.84 THz for both the harmonic trial and the AIMD-TDEP limit, but 6.11 THz for the effective FC2. The two targets differ in two ways: classical nuclei rather than quantum mode amplitudes, and the actual anharmonic displacement distribution rather than a Gaussian one. To separate them, 40 configurations (20 ± pairs) were drawn from the same harmonic trial with classical amplitudes $\sigma_e^2 = k_B T/\omega_e^2$ and computed with DFT. Their effective FC2 lies 0.08 THz from the quantum one (iteration-0 pool, mean frequency 6.05 against 6.11 THz; rms displacements 0.153 and 0.158 Å) but 0.23 THz from the AIMD-TDEP limit (rms displacement 0.154 Å). At 300 K in Al the difference is therefore mostly not quantum. It comes from sampling the actual, non-Gaussian distribution instead of a Gaussian centered on the trial, and the least-squares fit to that distribution gives an FC2 close to the harmonic one. AIMD-TDEP and the self-consistent protocols therefore converge to different quantities, and the force-call comparison applies to each method's own target.

Supplementary Fig. S4 shows, branch by branch, how the SOTC-D dispersions of Al and Au converge with 2–16 force calls; for Au, whose single-configuration SOTC-C fits are unstable, SOTC-D from the updated trial is stable at every budget. Figure 3 shows the resulting dispersions for six representative crystals. The difference between the effective FC2 and the large-cell harmonic reference combines the thermal renormalization with the truncation of the SOTC cell, which Table III separates. The renormalization is largest for the optical branches of the polar crystals. In PbTe at 700 K, the zone-center transverse optical mode, which is soft in the harmonic model (1.1 THz in the 128-atom cell), is stiffened to about 2.5 THz, the anharmonic hardening of the soft mode of this incipient ferroelectric. In MgO at 1500 K the optical branches differ by up to 3 THz, of which 1.4 THz (rms) is renormalization in the same cell. Si softens slightly at 600 K (the zone-center optical mode from 15.4 to 15.0 THz), whereas Mg stiffens (the highest zone-center mode from 7.2 to 7.6 THz). The metals Al and W change little. In all six crystals, SOTC-D with eight force calls follows the 40-snapshot effective dispersion across the Brillouin zone, for the iteration-1 design and for the bootstrapped design started from the harmonic trial, including the soft optical branch of PbTe; the largest deviations are in the optical branches of MgO and PbTe near X and W.

E. Force-constant basis and regression frameworks

Because the information matrix is fixed by the data and the basis, the basis is the other lever on the variance. Table VII compares the supercell basis with the crystal-symmetry basis. The latter has 3 (fcc), 7 (bcc), or 8 (rocksalt) parameters instead of 30–90, yet the held-out force errors of the two 40-snapshot effective models agree within 0.007 eV/Å, and their dispersions within 0.04–0.18 THz; for Si and Mg the SOTC cells carry the full point symmetry and the two bases coincide. With fewer parameters every design converges faster, so the absolute errors of Table S2 are lower, and the ranking of the designs is preserved: in the crystal basis, SOTC-D with the updated trial is better than canonical sampling in 8, 9, and 9 of 11 crystals at two, four, and eight calls, and bootstrapped SOTC-D beats one-shot canonical pairs in 7, 9, 9, and 10 of 11 at 2–16 calls (Tables S2 and S8). The margins are smaller than in the supercell basis, because with three to eight parameters even canonical designs are well conditioned.

The regression method matters much less than either. Fitting the same data with the hiPhive cluster space[21] and nine methods of the trainstation library distributed with hiPhive (least squares, ridge, Bayesian ridge, automatic relevance determination, recursive feature elimination, orthogonal matching pursuit, LASSO, adaptive LASSO, elastic net) changes the dispersion error at eight calls by at most 0.03 THz for any material and design (Table S4), and least squares is never significantly worse than a sparse or Bayesian method; with SOTC-D data the error is lower than with canonical data for every method in 9 of the 11 crystals. Regularization and sparsity are therefore no substitute for the design in this regime, in which the parameter count is small and the designs are full rank.

F. Other structure classes: diamond Si and hcp Mg

To test transferability beyond the cubic metals and rocksalt compounds, the complete protocol was applied without changes to covalent diamond Si and to hexagonal Mg. The same-cell harmonic FD models lie within 0.07 (Si) and 0.05 THz (Mg) of the 128- and 150-atom references, and thermal renormalization at 600 K changes the dispersions by 0.26 and 0.28 THz (Table III), softening Si and stiffening Mg. SOTC-D with eight force calls re-

TABLE V. Updated trial (iteration 1, supercell basis). Mesh RMSE (THz) against the iteration-1 effective FC2. Matched trial: SOTC-D designed from the iteration-1 trial (D_1) versus the better of random canonical subsets and pairs drawn from the same trial (C_1), scored against the disjoint effective FC2. Equal cost: bootstrapped SOTC-D (B; first pair from the harmonic trial, later pairs from the model refitted to the first pair, no extra force calls) and one-shot SOTC-D (D_0) versus canonical pairs from the harmonic trial (C_0; a single sTDEP iteration), scored against the 40-snapshot iteration-1 effective FC2, which shares no configuration with them. B and D_0 coincide at 2 calls by construction. The best entry of each group is bold.

	matched, 2		equal cost, 2			matched, 8		equal cost, 8		
	D_1	C_1	B	D_0	C_0	D_1	C_1	B	D_0	C_0
Al	**0.12**	0.30	**0.15**	0.15	0.31	**0.08**	0.13	**0.07**	0.09	0.12
Ag	**0.14**	0.23	**0.12**	0.12	0.32	**0.11**	0.12	**0.09**	0.13	0.16
Au	**0.12**	0.18	**0.24**	0.24	0.56	**0.07**	0.11	**0.07**	0.20	0.25
Cu	2.85	**2.12**	**2.85**	2.85	3.55	**0.18**	0.28	0.31	**0.28**	0.39
Mo	**0.12**	0.17	**0.10**	0.10	0.17	**0.06**	0.07	0.06	**0.04**	0.08
W	**0.09**	0.13	**0.08**	0.08	0.12	**0.04**	0.05	0.05	**0.04**	0.05
NaCl	**0.13**	0.29	**0.21**	0.21	0.41	**0.11**	0.13	**0.10**	0.18	0.18
MgO	**0.34**	0.60	**0.54**	0.54	0.68	**0.20**	0.29	**0.19**	0.38	0.36
PbTe	**0.08**	0.16	**0.19**	0.19	0.20	**0.04**	0.08	**0.06**	0.13	0.12
Si	**0.07**	0.10	**0.10**	0.10	0.11	**0.05**	0.05	**0.05**	0.09	0.07
Mg	**0.05**	0.06	**0.06**	0.06	0.07	**0.03**	0.05	**0.02**	0.06	0.04
$D_1 < C_1$ / $B < C_0$	10/11		11/11			11/11		11/11		
held-out force $D_1 < C_1$	11/11					10/11				

TABLE VI. Self-consistent protocols with DFT forces, equal cost. Mesh RMSE (THz) against the 40-snapshot iteration-1 effective FC2 (supercell basis). B: bootstrapped SOTC-D. sTDEP: half of the calls as canonical ± pairs from the harmonic trial, half as canonical ± pairs from the model refitted to them, fitted to the second iteration only (last) or to both (pool); median of three seeds. C_0: one-shot canonical pairs from the harmonic trial (median of 20 subsets). No configuration of any protocol enters the reference. The best entry of each group is bold.

	8 calls				16 calls			
	B	sTDEP last	sTDEP pool	C_0	B	sTDEP last	sTDEP pool	C_0
Al	**0.07**	0.15	0.12	0.12	**0.06**	0.10	0.08	0.10
Ag	**0.09**	0.12	0.14	0.16	**0.08**	0.10	0.12	0.13
Au	**0.07**	0.13	0.27	0.25	**0.06**	0.09	0.19	0.24
Cu	**0.31**	0.60	0.41	0.39	0.25	0.28	**0.18**	0.29
Mo	**0.06**	0.12	0.07	0.08	0.05	0.07	**0.05**	0.07
W	**0.05**	0.08	0.05	0.05	**0.03**	0.05	0.04	0.04
NaCl	**0.10**	0.20	0.14	0.18	**0.07**	0.16	0.18	0.15
MgO	**0.19**	0.32	0.33	0.36	**0.16**	0.30	0.26	0.30
PbTe	**0.06**	0.13	0.09	0.12	**0.06**	0.09	0.08	0.10
Si	**0.05**	0.08	0.08	0.07	**0.04**	0.05	0.06	0.06
Mg	**0.02**	0.05	0.03	0.04	**0.02**	0.04	0.04	0.03
B best	11/11				9/11			

produces the effective FC2 within 0.04 (Si) and 0.03 THz (Mg), the same accuracy as for the bcc metals. It is the best design at two calls for both crystals (Si: 0.07 versus 0.10 THz for paired canonical sampling; Mg: 0.07 versus 0.08) and equal to paired canonical sampling at eight; special displacements and rattle are again worse (Table IV), and bootstrapped SOTC-D reaches 0.05 (Si) and 0.02 THz (Mg) at eight calls (Table V). The behavior of the method thus does not depend on the structure class.

G. Higher-order force constants

The constrained regression also extends to anharmonic tensors when enough forces are available. Joint FC2+FC3+FC4 fits to 80 canonical snapshots in $3\times3\times3$ cells (Al 300 K, W 1500 K, NaCl 700 K) lower the held-out force RMSE relative to the harmonic FD FC2 by 93.7%, 87.5%, and 72.3%, with full-rank designs (106, 126, and 48 parameters) and held-out R^2 of 0.9997, 0.9994, and 0.969 (Fig. S1). These tensors define the anharmonic model potentials (surrogates) of the next subsection. They are force-regression results; lifetimes and thermal conductivity are not computed.

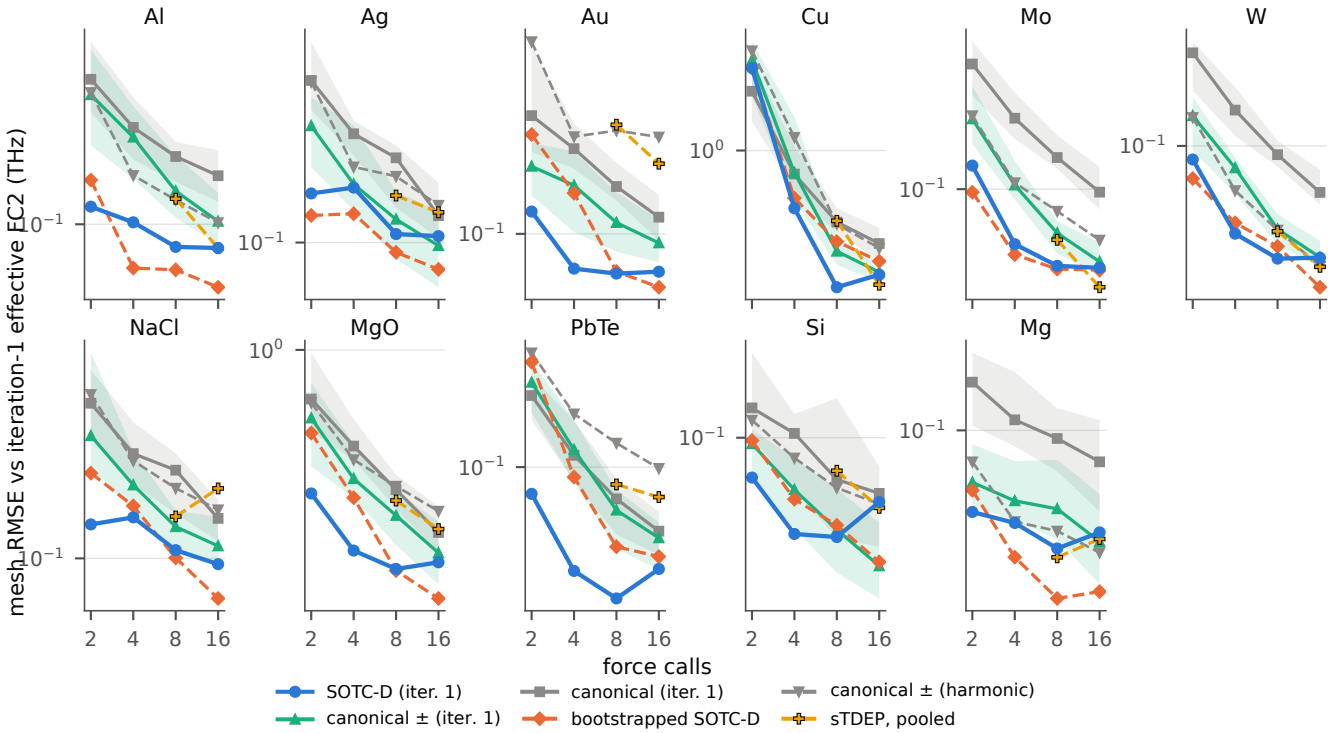


FIG. 2. Mesh RMSE against the iteration-1 effective FC2 versus force calls for the eleven crystals. Solid lines: matched updated trial, SOTC-D (blue) versus random canonical pairs (green) and subsets (gray), scored against the disjoint effective FC2 (bands: 10–90% over 20 draws). Dashed lines: equal cost from the harmonic trial, bootstrapped SOTC-D (orange) versus one-shot canonical pairs (gray) and two-iteration sTDEP fitted to both iterations (yellow, 8 and 16 calls, median of three seeds), scored against the full iteration-1 effective FC2, with which they share no configuration.

TABLE VII. Supercell versus crystal space-group FC2 basis: number of parameters p, alias-free crystal-basis cutoff r_c, held-out force RMSE (eV/Å) of the 40-snapshot iteration-1 effective FC2, and mesh RMSE (THz) between the two effective FC2s.

	p_{sc}	p_{cr}	r_c (Å)	F_{sc}	F_{cr}	$\Delta\omega$
Al	30	3	3.78	0.089	0.092	0.18
Ag	30	3	3.84	0.255	0.257	0.07
Au	30	3	3.85	0.340	0.347	0.11
Cu	90	3	3.42	0.320	0.319	0.13
Mo	78	7	4.66	0.274	0.276	0.07
W	78	7	4.69	0.283	0.283	0.04
NaCl	53	8	4.31	0.223	0.223	0.05
MgO	53	8	3.21	0.504	0.504	0.14
PbTe	53	8	5.01	0.259	0.259	0.04
Si	10	10	5.01	0.191	0.191	0.00
Mg	12	12	4.61	0.146	0.146	0.00

H. Anharmonic model potentials with known self-consistent solutions

With DFT forces the self-consistent effective FC2 of Eq. (3) is not known exactly, and the comparisons above are made against finite-sample estimates. Anharmonic model potentials fitted to DFT data remove this limitation: their forces are noise-free and cheap, so the self-consistent solution can be converged to any precision and each protocol can be repeated many times. To separate the ingredients of SOTC-D, the FC2+FC3+FC4 models fitted to 80 QE snapshots of Al, W, and NaCl (Sec. V G) were used as noise-free anharmonic surrogates, with the effective FC2 converged on 6000 canonical samples (0.001–0.006 THz between 3000 and 6000). Thirty repetitions of each design were scored (Table S1 in the Supplementary Material). Antithetic pairing is the largest single ingredient: it lowers the median mesh error of two-configuration canonical designs from 0.199 to 0.066 THz (Al), 0.091 to 0.035 THz (W), and 0.224 to 0.176 THz (NaCl), because the cubic force, which is pure noise for an FC2 fit, cancels exactly. Exact r.m.s. amplitudes improve the larger budgets further (16 calls: 0.025 versus 0.032 THz for Al and 0.063 versus 0.078 THz for NaCl). D-optimal signs add little in these small surrogate cells, which have only 7–8 FC2 parameters. In the 30–90-parameter SOTC cells the signs matter more: special displacements, which fix the amplitudes but choose the signs by orthogonal Walsh patterns rather than by D-optimality, are worse than SOTC-D in every crystal at eight calls (Sec. V C). In these one-shot designs with up to 16 calls, stratified Gaussian-quantile amplitudes, which match the fourth moment of the canonical distri-

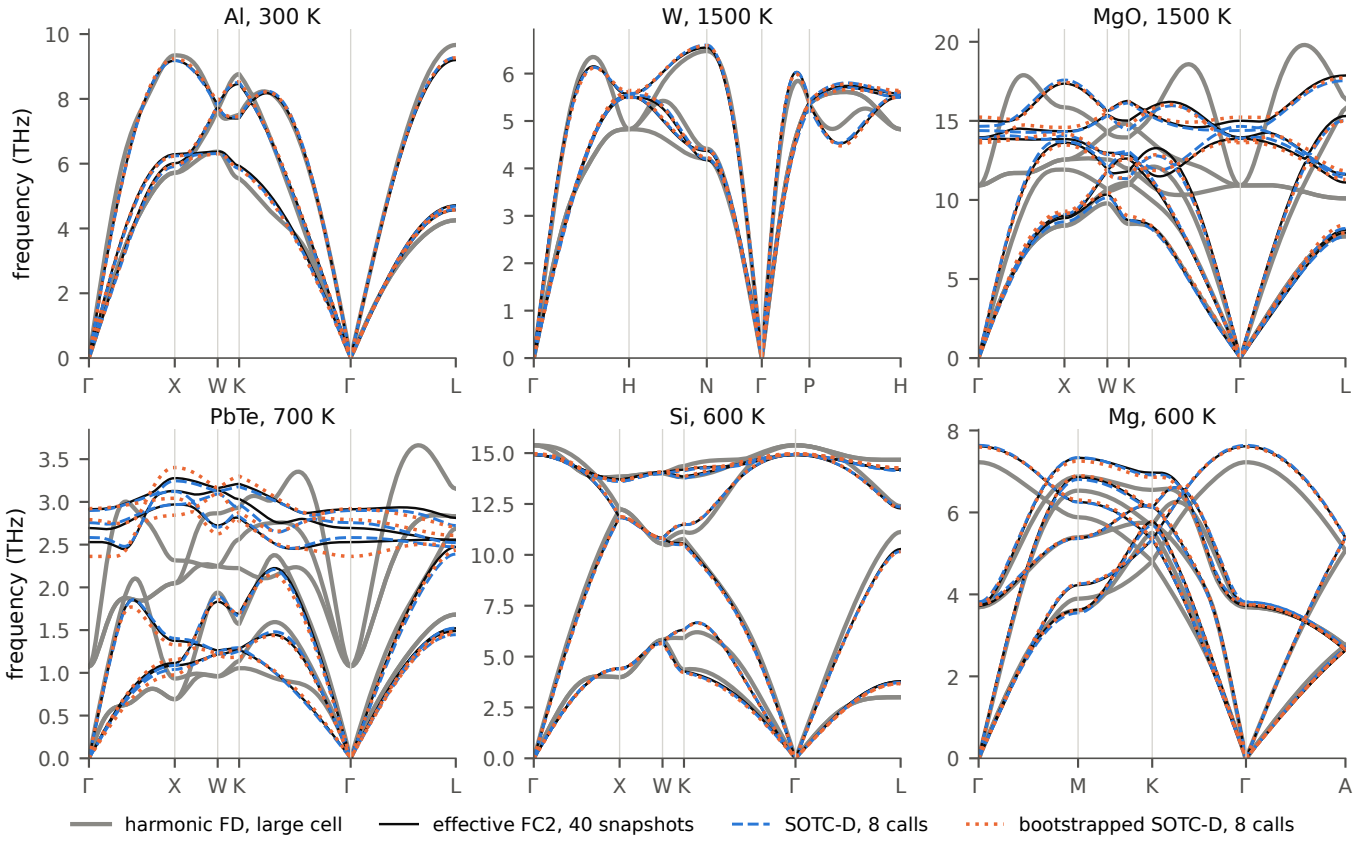


FIG. 3. Dispersions of the converged large-cell harmonic FD reference (gray), the 40-snapshot iteration-1 effective FC2 (black), and SOTC-D with eight force calls from the iteration-1 trial (blue dashed) and bootstrapped from the harmonic trial (orange dotted), for six representative crystals; effective models are Fourier interpolated from the SOTC cell, and the non-analytic correction is omitted for MgO and PbTe.

bution, give no further improvement, and no amplitude bias of the fixed-amplitude design is detectable, as expected from Eq. (8).

Self-consistent protocols and SSCHA. The surrogates also give the exact self-consistent target, obtained by iterating the canonical fit with 6000 samples per iteration to a mesh change below 0.01 THz, so that complete self-consistent workflows can be compared at equal numbers of force calls (Table VIII): one-shot canonical pairs and SOTC-D from the harmonic trial; sTDEP with three iterations of equal size, fitted to the last iteration or to all of them; SOTC-D-SC (Sec. II F); and SSCHA, run with python-sscha[6] with the surrogate as an exact calculator, starting from the same harmonic trial (minimization step 0.1, threshold 0.5 on the Kong–Liu effective sample size, stopping when the gradient is below 0.2 times its statistical error, fixed centroids, at most 30 populations of $N = 8$–128 configurations, 5 repetitions against 12 for the other protocols). Because the smallest SSCHA population has eight configurations, a factor of 2 between SSCHA and SOTC-D-SC at a threshold reached with four calls is limited by this grid. For Al and W, whose harmonic trials are within 0.15–0.22 THz of the self-consistent target, all protocols are comparable; SOTC-D-SC is best at 16 and 32 calls and within 0.003 THz of the best at 8 calls. NaCl is the stringent case: the harmonic trial is 0.53 THz from the self-consistent target, so one-shot designs stall at 0.27–0.33 THz irrespective of the budget. SOTC-D-SC reaches 0.096, 0.054, 0.035, and 0.029 THz with 8, 16, 32, and 64 calls, whereas sTDEP fluctuates between 0.036 and 0.18 THz because each iteration redraws a small sample, and SSCHA needs 96 calls for 0.041 THz and 192 for 0.027 THz. At equal accuracy, SOTC-D-SC therefore needs about three times fewer force calls than SSCHA for NaCl, and it is the most accurate protocol at 32 and 64 calls; only pooled sTDEP is marginally better at 16 calls (0.048 versus 0.054 THz).

The same study was repeated for diamond Si and hcp Mg at 600 K, with FC2+FC3+FC4 surrogates fitted to all 194 same-cell QE configurations of each crystal (cutoffs 5.0/4.0/2.5 Å for Si and 4.6/3.5/3.3 Å for Mg; held-out force RMSE 0.075 and 0.040 eV/Å, $R^2 = 0.993$ and 0.992, against 0.95 and 0.88 for FC2 alone). Their harmonic trials are 0.12 and 0.27 THz from the self-consistent target. SOTC-D-SC reaches 0.037, 0.024, 0.014, and 0.013 THz for Si and 0.026, 0.021, 0.016, and 0.014 THz for Mg with 8, 16, 32, and 64 calls; pooled sTDEP is equal for Si and slightly better for Mg at 32

and 64 calls (0.015 and 0.011 THz), as is one-shot canonical sampling for Mg at 64 calls (0.010 THz), because the Mg trial changes little between iterations. SSCHA needs 64 (Si) and 16 calls (Mg) for 0.05 THz and 128 and 64 calls for 0.03 THz, against 4 and 16 (Si) and 4 and 8 calls (Mg) for SOTC-D-SC (Table X).

Amplitude bias. The flattening of the SOTC-D-SC error between 32 and 64 calls for Si and Mg is consistent with the fixed-amplitude bias of Eq. (8). It can be removed exactly at fourth order: if, for every mode, the squared amplitudes q_e^2/σ_e^2 are drawn without replacement within blocks of four pairs from the equal-weight values $\{0, 0, 2-\sqrt{2}, 2+\sqrt{2}\}$, independently for each mode, they have the Gaussian moments $\langle q_e^2\rangle = \sigma_e^2$ and $\langle q_e^4\rangle = 3\sigma_e^4$, so that the V_{aeee} term of Eq. (8) vanishes while the signs remain D-optimal. On the five surrogates with 128 calls (12 repetitions), this moment-matched variant lowers the error from 0.0098 to 0.0080 (Al), 0.0080 to 0.0047 (W), and 0.0122 to 0.0092 THz (Mg) and raises it from 0.0074 to 0.0092 (Si) and 0.0170 to 0.0181 THz (NaCl); with 16–32 calls the two variants differ by 0.001–0.015 THz in either direction. The fixed-amplitude bias is therefore of order 0.01 THz, an order of magnitude below the supercell and reference errors of Sec. V B, and fixed amplitudes, which minimize the variance at small budgets, remain the default; the moment-matched variant is the appropriate choice for budgets above about 100 calls.

I. Cost

Table IX converts the comparison into computing time. A force call in the SOTC cell costs 0.3–6.9 core-hours; the harmonic FD trial costs three to nine calls. To reach the iteration-1 dispersion error that SOTC-D attains with two calls, paired canonical sampling needs 4–16 calls, and to reach its eight-call error, more than 16 calls for 9 of the 11 crystals.

For complete self-consistent workflows, Table X gives the number of force calls that each complete protocol, started from the same harmonic trial, needs to reach and keep a target accuracy on the model potentials of Sec. V H. SOTC-D-SC needs the fewest calls, or is tied for the fewest, in 13 of the 14 attainable material–threshold combinations of the five surrogates (the exception is Mg at 0.02 THz, 32 versus 16 calls for one-shot canonical pairs), and strictly the fewest for W at 0.02 THz and for NaCl at 0.05 and 0.03 THz, where it needs 32 and 64 calls against 96 and 192 for SSCHA, 64 or more for sTDEP, and more than 64 for every one-shot protocol. It needs a factor of 2–16 fewer calls than SSCHA in 13 of the 14 combinations and as many in the remaining one (Al, 0.03 THz). Since all protocols evaluate forces in the same cell, the cost in core-hours is proportional to these numbers, with the per-call costs of Table IX.

J. Thermodynamics and displacement parameters

Because C_V is close to the Dulong–Petit limit at T_{des} for all nine materials, entropy, free energy, and Debye temperature are the more discriminating thermodynamic quantities (Table XI). The SOTC-C models reproduce the effective S_{vib} to within 0.2–1.4 $\mathrm{J\,mol^{-1}K^{-1}}$ for the metals, NaCl, and MgO (2.3 for PbTe) and the effective T_D to within 19 K, whereas the harmonic references of Ag, Au, Cu, NaCl, MgO, and PbTe are higher in S_{vib} by up to 5.3 $\mathrm{J\,mol^{-1}K^{-1}}$ (Au) at the same volume; for Al, Mo, and W the harmonic and effective values agree within 0.4 $\mathrm{J\,mol^{-1}K^{-1}}$. DOS-integrated quantities are thus considerably less sensitive to the design than branch-resolved frequencies. Using the PBE equilibrium lattice parameters (0.01–1.1% from the production values) changes the harmonic S_{vib} at T_{des} by up to 1.7 $\mathrm{J\,mol^{-1}K^{-1}}$ (Cu, PbTe) and the mesh frequencies by 0.00–0.35 THz (Supplementary Table S6).

For the Debye–Waller comparison the force constants must correspond to the measurement temperature and volume. The converged harmonic models at the room-temperature lattice parameters give $U_{\mathrm{iso}}(300\,\mathrm{K})$ of 0.0104 (Al), 0.0066 (Cu), and 0.0081 Å^2 (Au), within 6% of the values 0.0099, 0.0070, and 0.0078 Å^2 tabulated by Peng *et al.* at 293 K[30]. For Al, whose production lattice parameter is the room-temperature one, the SOTC-cell models can be compared directly (Fourier-interpolated on a 20^3 mesh). The 300 K effective FC2 gives 0.0104 Å^2, and the SOTC-D models with eight force calls give 0.0102 (harmonic trial), 0.0104 (bootstrapped), and 0.0106 Å^2 (updated trial), 3–7% above the tabulated value. The harmonic FD model of the same 32-atom cell gives 0.0150 Å^2, because its interpolated spectrum has small regions of imaginary and near-zero frequencies (minimum −0.67 THz). At the production lattice parameters, Ag, Mo, and W give 0.0096, 0.0029, and 0.0022 Å^2, 3–10% above the tabulated 0.0093, 0.0027, and 0.0020 Å^2, as expected because the production lattice parameters of these crystals are those near their design temperatures (1000–1500 K) and thus larger than at room temperature. Debye–Waller factors at the design temperatures require force constants and lattice parameters at those temperatures; they are not compared with room-temperature data.

K. Thermal expansion of Al

Because the SCHA free energy is stationary with respect to the auxiliary force constants, its temperature derivative at fixed volume is the harmonic entropy of the self-consistent FC2, $S = S_{\mathrm{h}}[\mathbf{\Phi}_{\mathrm{SC}}](T)$. The free energy per atom on the volume grid is

$$F(V,T) = E_{\mathrm{st}}(V) + F_{\mathrm{h}}[\mathbf{\Phi}_{\mathrm{SC}}(V,T)](T) + \left\langle E - E_0 - \tfrac{1}{2}\mathbf{u}\cdot\mathbf{\Phi}_{\mathrm{SC}}\mathbf{u}\right\rangle, \qquad (11)$$

TABLE VIII. Self-consistent protocols on the anharmonic surrogates. Median mesh RMSE (THz) against the self-consistent effective FC2 (= SCHA auxiliary FC2) at 8/16/32/64 force calls (12 repetitions). SSCHA: python-sscha with the surrogate as calculator, populations of $N = 8$–128 configurations until convergence (5 repetitions), listed at the median total number of force calls (the better population size where two need the same number). Other SSCHA budgets: Al 128 calls, 0.008; W 128 calls, 0.009; NaCl 96 calls, 0.041; 192 calls, 0.027; 384 calls, 0.025; Si 128 calls, 0.024; 256 calls, 0.020; Mg 128 calls, 0.015 THz.

Protocol	Al	W	NaCl	Si	Mg
Canonical ±, one shot	.042/.030/.026/.019	.029/.020/.020/.017	.271/.278/.299/.331	.036/.023/.025/.021	.026/.019/.018/.010
SOTC-D, one shot	.031/.028/.017/.013	.023/.020/.018/.013	.273/.308/.312/.309	.032/.025/.018/.023	.027/.023/.021/.018
sTDEP (last iteration)	.051/.038/.039/.026	.030/.026/.021/.013	.150/.055/.091/.036	.049/.035/.029/.019	.040/.028/.024/.017
sTDEP (pooled)	.044/.040/.019/.013	.024/.017/.014/.008	.181/.048/.060/.081	.038/.024/.016/.014	.033/.022/.015/.011
SOTC-D-SC	.034/.022/.016/.012	.020/.014/.011/.008	.096/.054/.035/.029	.037/.024/.014/.013	.026/.021/.016/.014
SSCHA	.034/.030/.021/.010	.023/.017/.012/.016	–/.092/.069/–	.098/.070/–/.042	.055/.041/–/.019
Harmonic vs self-consistent	0.215	0.151	0.528	0.121	0.268

TABLE IX. Cost. t: measured cost per force call in the SOTC cell (core-hours, 32 cores, mean over all same-cell calls); N_{FD}: calls of the harmonic FD trial. Columns 4–5: number of random canonical pairs (force calls) from the same trial needed to reach the iteration-1 mesh error that SOTC-D attains with 2 and 8 calls.

	t (core-h)	N_{FD}	match $\mathrm{D}_1(2)$	match $\mathrm{D}_1(8)$
Al	0.28	5	16	>16
Ag	1.06	5	8	16
Au	1.09	5	8	>16
Cu	0.58	7	4	>16
Mo	6.75	7	4	>16
W	6.92	7	4	>16
NaCl	1.28	9	8	>16
MgO	0.47	9	8	>16
PbTe	3.83	9	8	>16
Si	0.39	3	4	16
Mg	1.13	3	16	>16

where E_{st} is the static energy (third-order Birch–Murnaghan fit of the primitive-cell PBE energies), F_{h} is the quantum harmonic free energy, and the last term, averaged over the last two designed pairs, is the anharmonic energy correction of the SCHA[4,6]. The linear thermal expansion coefficient follows as $\alpha = (\partial S/\partial V)_T/3B_T$, with the isothermal bulk modulus $B_T = V\partial^2 F/\partial V^2$, both at the volume that minimizes F. The quasiharmonic approximation (QHA) with the harmonic FD FC2 in the same cell uses the FD FC2 and no correction term. Neither requires derivatives of the FC2 with respect to temperature.

Table XII gives the results. The QHA with the converged 125-atom harmonic FC2 gives 24.1 and 26.2 × 10^{-6} K^{-1} at 300 and 400 K, 4% above the measured 23.2 and 25.1, and 40.2 at 900 K, against 36 at 923 K, of which vacancies contribute about two units[31]. With the harmonic FD FC2 of the 32-atom cell, the QHA overestimates α by 13–15% at 300–400 K and diverges above 500 K; at 900 K its free-energy minimum lies outside the volume grid. The cause is the truncated cell: its interpolated harmonic spectrum has imaginary frequencies at all but the smallest volume, with the minimum frequency (full 20^3 mesh) falling from −0.45 to −1.25 THz between 4.025 and 4.150 Å. The self-consistent FC2 is stable at every volume and temperature (minimum frequency 0.30–0.40 THz).

Two independent self-consistent grids were computed, one with design seeds that differ between volumes and one with the same seeds at all volumes. They agree within 0.2×10^{-6} K^{-1} at 100 K, 1.5×10^{-6} K^{-1} at 300 K, and 3×10^{-6} K^{-1} at 400–500 K. At 300 and 400 K they lie 0–7% and 5–17% below experiment, and below the converged QHA. Above 500 K the two grids differ by up to 10×10^{-6} K^{-1}, so the self-consistent thermal expansion is not determined there. The reason is the curvature of $F(V)$. Across the grid, F varies by a few meV per atom, and the statistical error of F_{h} from an FC2 fitted to 16 configurations (about 0.05 THz in the frequencies) is a sizable fraction of that at high temperature. The anharmonic energy term (between −0.4 and −17 meV per atom) is not the main source: omitting it changes α by less than the difference between the grids. The fixed amplitudes of the design bias the energy term as they bias the FC2 [Eq. (8)], which is a further, smaller error at high temperature.

In summary, the self-consistent design removes the instability of the small-cell QHA and gives the thermal expansion of Al within 17% of experiment at 300–400 K, reproducibly up to 500 K. Close to melting, the free-energy curvature needs many more configurations per volume than the budgets for which SOTC-D was designed. For a crystal whose large-cell harmonic FC2 is stable, such as Al, the converged harmonic QHA is the more accurate route. The calculation used 1379 force calls in the 32-atom cell (two grids of seven volumes and six temperatures with 16 calls each, plus the FD trials) and 21 in the 125-atom cell.

L. Polar crystals

Born effective charges and dielectric constants from DFPT at the production volumes are $(Z^*_{\mathrm{cation}}, \varepsilon_\infty) = (1.111, 2.471)$ for NaCl, $(1.973, 3.167)$ for MgO, and $(5.767, 28.83)$ for PbTe. Applied to the selected SOTC-C

TABLE X. End-to-end cost on the anharmonic surrogates: force calls needed by each complete protocol, started from the same harmonic trial, to reach and keep a median mesh error of 0.05, 0.03, 0.02 THz against the self-consistent effective FC2 (tested budgets 4–64 calls; SSCHA 8–384).

Protocol	Al	W	NaCl	Si	Mg
Canonical ±, one shot	4/32/64	4/8/16	>64/>64/>64	4/16/>64	4/8/16
SOTC-D, one shot	8/16/32	4/4/16	>64/>64/>64	4/16/>64	4/8/64
sTDEP (last)	16/64/>64	8/8/64	64/>64/>64	8/32/64	8/16/64
sTDEP (pooled)	4/32/32	4/4/16	>64/>64/>64	4/16/32	4/16/32
SOTC-D-SC	4/16/32	4/4/8	32/64/>64	4/16/32	4/8/32
SSCHA	8/16/64	8/8/16	96/192/>384	64/128/>256	16/64/64

TABLE XI. Vibrational thermodynamics per atom at T_{des} and displacement parameters. S_{vib} ($\mathrm{J\,mol^{-1}K^{-1}}$), F_{vib} ($\mathrm{kJ\,mol^{-1}}$), and T_D (K) from the converged harmonic FD (FD), the same-cell effective FC2 (eff.), and the SOTC-C model. $U_{\mathrm{iso}}(300\,\mathrm{K})$ ($\mathrm{Å}^2$) from the converged harmonic model at the room-temperature (Al, Cu, Au) or production lattice, compared with Peng *et al.*[30] at 293 K.

	S_{vib}			F_{vib}			T_D			$U_{\mathrm{iso}}(300\,\mathrm{K})$	
	FD	eff.	SOTC-C	FD	eff.	SOTC-C	FD	eff.	SOTC-C	calc.	Ref.[30]
Al	27.8	27.8	28.2	-0.2	-0.2	-0.3	397	394	396	0.0104	0.0099
Ag	72.9	69.8	68.9	-47.9	-44.8	-43.9	210	238	252	0.0096	0.0093
Au	84.1	78.8	78.0	-70.9	-64.6	-63.8	168	204	204	0.0081	0.0078
Cu	68.7	65.9	65.7	-52.5	-49.0	-48.8	292	332	335	0.0066	0.0070
Mo	68.4	68.8	67.4	-65.1	-65.7	-63.6	358	354	373	0.0029	0.0027
W	72.8	72.7	73.1	-71.7	-71.6	-72.2	300	303	300	0.0022	0.0020
NaCl	58.8	55.9	56.7	-23.6	-21.5	-22.1	254	290	286	0.0224	–
MgO	51.8	49.8	50.1	-39.8	-36.7	-37.3	726	780	774	0.0044	–
PbTe	76.4	75.2	72.9	-36.0	-35.2	-33.5	133	140	155	0.0159	–

TABLE XII. Linear thermal expansion coefficient α (10^{-6} $\mathrm{K^{-1}}$) of Al (PBE), from $(\partial S/\partial V)_T/3B_T$ on seven volumes. SC: self-consistent SOTC-D in the 32-atom cell with 16 force calls per volume and temperature, from two independent grids (design seeds independent across volumes, and common to all volumes). QHA: harmonic FD FC2 in the same cell and in the converged 125-atom cell. Experiment: averages of measured values at 298, 400, and 923 K (300, 400, and 900 K rows) compiled in Ref. 31. a: lattice parameter (Å) of the converged QHA.

	SC		QHA			
T (K)	indep.	common	32 at.	125 at.	expt.	a
100	13.1	12.9	14.6	12.9	–	4.055
300	21.7	23.2	26.3	24.1	23.2	4.072
400	23.9	20.9	28.9	26.2	25.1	4.082
500	24.8	24.8	31.8	28.2	–	4.093
700	19.9	28.9	41.1	32.9	–	4.118
900	22.4	12.4	79.4[a]	40.2	36.0	4.148

[a] Minimum outside the volume grid.

refits, they open zone-center LO–TO splittings of 2.29, 8.05, and 1.32 THz (Fig. 4). For NaCl, the same correction applied to the $3 \times 3 \times 3$ harmonic FD model gives TO and LO limits of 4.37 and 7.26 THz, 11% and 10% below the infrared values of 4.92 and 8.03 THz[32]; the LO/TO ratio, 1.66, is within 2% of experiment. The non-analytic correction is independent of the displacement design and can be combined with any of the FC2 models above (NaCl comparison with neutron and infrared data in Supplementary Fig. S3).

VI. DISCUSSION

The results allow a precise statement of what correlator design contributes. The correlator objective of SOTC-C matches shell-averaged second moments, which by construction the canonical distribution also matches on average; it does not control the per-mode amplitudes or the conditioning of the force fit, and its regression adds an axial prior that is inexact for fcc first-neighbor bonds. Consequently the SOTC-C models are good effective-FC2 estimates—their thermodynamics agree closely with the effective references—but their dispersions are not systematically better than those of canonical sampling. SOTC-D keeps the deterministic, temperature-aware character of SOTC but optimizes what matters for regression: exact thermal mode variances in every configuration, a maximal information determinant, and cancellation of the cubic force by antithetic pairs.

Within the class of finite-temperature effective-harmonic methods that fit force constants to sampled forces, the comparisons above cover the relevant alternatives at equal cost and with identical data: canonical sampling, unpaired and paired, fitted by least squares as in TDEP[8,9], and its iterated form sTDEP[12]; SSCHA[4,6], whose fixed point is the same self-consistent effective FC2; special displacements[15]; thermal rattle

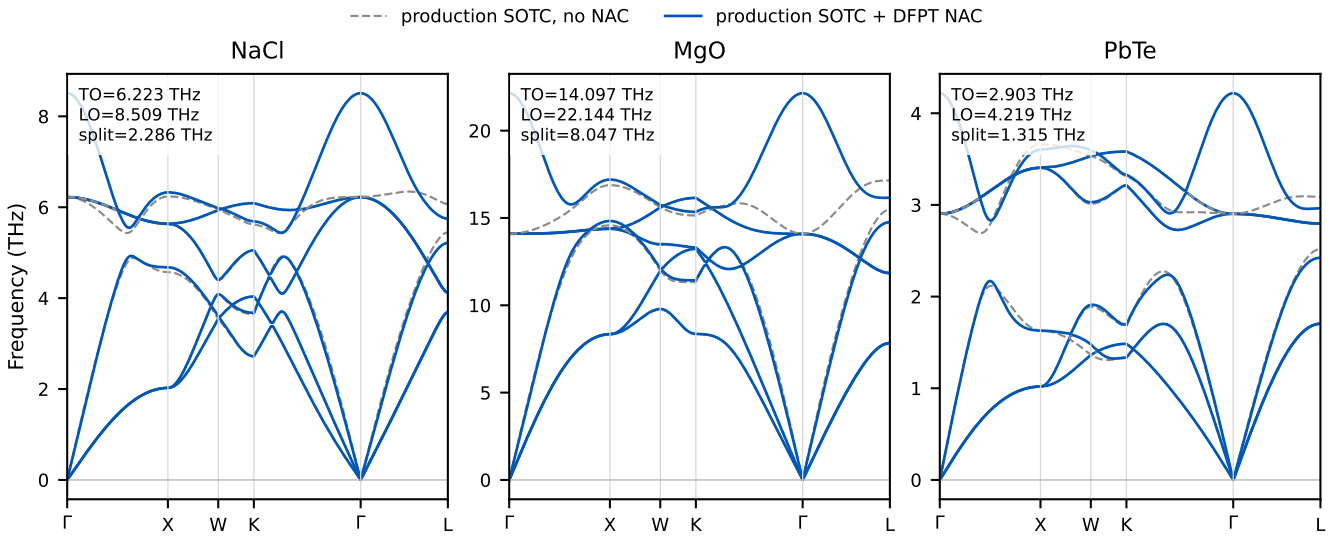


FIG. 4. SOTC-C FC2 dispersions of NaCl, MgO, and PbTe without (gray dashed) and with (blue) the non-analytic correction from same-volume QE-DFPT Born charges and dielectric tensors. Zone-center TO, LO, and splitting values are given in each panel.

with hiPhive regression[21]; and nine sparse, Bayesian, and least-squares fit methods. Three results stand out. First, at the small budgets typical of high-throughput or expensive-functional calculations, SOTC-D is the best design with the same trial in 10 of 11 crystals at two and at eight force calls (8 of 11 at four), and its bootstrapped form beats one-shot canonical sampling in all 11 crystals at every budget from 4 to 16 calls and two-iteration sTDEP with DFT forces in all 11 at eight calls and 9 of 11 at sixteen. Second, on model potentials with an exactly known self-consistent target, SOTC-D-SC is at least as accurate as sTDEP and SSCHA in 50 of 56 comparisons at equal budget; in the other six, pooled sTDEP (W and NaCl at 64 and 16 calls, respectively, Si at 16, Mg at 32 and 64) or SSCHA (Al at 64) is better by at most 0.006 THz. It needs a factor of 2–16 fewer force calls than SSCHA for a given accuracy in 13 of 14 cases, and with DFT forces for Al it reaches with four calls the accuracy SSCHA reaches with 256 when both are measured against the iteration-1 effective FC2; against the SSCHA result itself the two are not distinguished at the 0.1 THz resolution of that test. Third, the regression method, including sparsity-promoting and Bayesian fits, changes the result by at most 0.03 THz, whereas the design changes it by up to a factor of 2.6 relative to random canonical subsets at two and eight calls in the supercell basis (Table IV), and by less in the crystal basis; the crystal-symmetry basis reduces the number of parameters by an order of magnitude at no loss in held-out accuracy and should be used by any method. The gain of SOTC-D diminishes with budget, as expected once the sampling error falls below the supercell and reference errors.

The estimator theory explains these observations. The effective FC2 is the Gaussian-averaged Hessian; SOTC-D estimates it with a bias that is a single diagonal term of the M-mode renormalization sum, $O(1/M)$, whereas antithetic pairs remove the cubic force, the dominant noise of every unpaired design, exactly; fixed amplitudes and D-optimal signs then minimize the remaining variance for the given number of calls. The theory also identifies the one failure mode found in the iteration-0 benchmark: a trial with unstable commensurate modes (PbTe) gives a defective target and underexcites the soft modes for any design. Updating the trial during the design repairs it at no extra cost, and the same update converges to the SCHA fixed point.

SOTC-D changes only how the configurations are chosen; the target, the regression, and the self-consistency condition are those of sTDEP and the SCHA. It is therefore complementary to these methods rather than an alternative to them. Within sTDEP it replaces the Gaussian draws at each iteration, and the bootstrapped and self-consistent variants are sTDEP iterations in which all earlier forces are retained. Its use within SSCHA is less direct, because the SSCHA reweighting of a population to a new trial requires the probability density of the configurations, which the deterministic design does not provide; the free energy and its gradients would have to be evaluated from the fitted effective model instead. AIMD-based TDEP samples the classical anharmonic ensemble, including non-Gaussian displacement statistics, and is not replaced by a Gaussian design; SOTC-D addresses the self-consistent harmonic level, at which quantum zero-point motion enters through the mode amplitudes. For Al at 300 K the two effective FC2 models differ by 0.28 THz, mostly because of the non-Gaussian distribution sampled by the MD rather than zero-point motion (Sec. V D). With DFT forces, AIMD-

TDEP needed about 500 MD steps per trajectory to come within 0.1 THz of its own limit, whereas SOTC-D-SC needed four force calls to reach the same error relative to its target (Sec. V D); the 250 equilibration steps are part of this difference, and successive MD steps are correlated. Given the uncertainty of both references, this is an order-of-magnitude comparison.

SOTC-D gives the effective FC2 at one volume. Repeated on a volume grid, it gives the free energy of the self-consistent harmonic model, because the designed configurations also provide the energies for its anharmonic correction. For Al, with 16 force calls per volume and temperature, this removes the divergence of the QHA of the same 32-atom cell and gives the thermal expansion within 17% of experiment at 300–400 K, reproducibly up to 500 K (Sec. V K). At higher temperatures the curvature of the free energy with respect to volume is not resolved at this budget. Thermal expansion is therefore an application in which the small-budget advantage of the design is limited: it needs free-energy differences between volumes to better than 1 meV per atom, and for crystals with a stable harmonic FC2 in a large cell the converged QHA is the cheaper and more accurate route.

The comparison also shows where the remaining errors are. In 32–72-atom cells the supercell truncation error (up to 0.5 THz) is comparable to or larger than the sampling error of a few SOTC-D configurations; larger cells and the non-analytic correction for polar crystals are therefore needed for quantitative dispersions, independently of the sampling scheme. The method assumes a unimodal displacement distribution; double-well systems near displacive transitions are outside its scope, as are anharmonic free energies, lifetimes, and thermal conductivity, which require the higher-order tensors of Sec. V G together with appropriate many-body theory.

VII. CONCLUSIONS

We have introduced special optimal thermal cells for finite-temperature effective-harmonic phonons and compared two design objectives. Correlator-matched cells (SOTC-C) of nine crystals identify their FC2 models completely in a space-group adapted basis and reproduce effective thermodynamics, but their dispersion accuracy is limited by an inexact regression prior and by the lack of control over mode amplitudes; for Au, single-configuration iterations give an unstable model. The D-optimal antithetic design SOTC-D, which excites every mode at its exact thermal amplitude, chooses signs that maximize the FC2 information, and cancels cubic forces by inversion pairs, has an amplitude bias of order $1/M$ at fourth order and no cubic noise. With eight force calls it reproduces same-cell effective FC2 models within 0.03–0.31 THz for eleven crystals of five structure classes (fcc and bcc metals, rocksalt, diamond, and hcp), and with an updated trial it is more accurate than canonical sampling with the same trial in all eleven at four and eight calls. Its bootstrapped form needs no additional calculations and, in the supercell basis, outperforms one-shot canonical sampling in every crystal at every budget from 4 to 16 calls and two-iteration sTDEP in every crystal at eight calls; on anharmonic model potentials of five crystals, the fully self-consistent form reaches a given accuracy with a factor of 2–16 fewer force calls than SSCHA in 13 of 14 cases. Special displacements, thermal rattle, and alternative regression methods do not close this gap. Converged references reveal that finite-temperature renormalization (up to 1.4 THz) and supercell truncation (up to 0.5 THz) are as large as or larger than the sampling errors, and room-temperature Debye–Waller factors of Al, Cu, and Au are recovered within 6% of tabulated values when force constants at the room-temperature lattice parameter are used. On a volume grid, the self-consistent form gives the thermal expansion of Al within 17% of experiment at 300–400 K, reproducibly up to 500 K, whereas the QHA of the same small cell diverges above 500 K; at higher temperatures 16 force calls per volume do not resolve the free-energy curvature, and for Al the converged harmonic QHA is more accurate. Because the gain is largest at budgets of a few force calls, the design is most useful where each calculation is expensive—hybrid functionals, large cells, or high-throughput screening of finite-temperature phonons—and it can be used as the sampling step of existing self-consistent workflows. The implementation, inputs, and all comparison data are openly available.

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AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts to disclose.

Author Contributions

Albert Linda: Conceptualization; Methodology; Software; Investigation; Formal analysis; Writing – original draft; Writing – review & editing.

SUPPLEMENTARY MATERIAL

See the Supplementary Material for the derivations of Eqs. (4) and (8) and of the averaged-Hessian identity, the gradient of Eq. (6), the surrogate design study, the design comparison in the crystal-symmetry basis, the regression-framework comparison, the individual SOTC-C iterations, the lattice-parameter sensitivity, an overview of the

reference accuracy, the electronic-convergence and cutoff tests of the higher-order fits, the NaCl comparison with neutron data, the branch-resolved convergence of SOTC-D for Al and Au with the Al neutron model, and the design comparisons at four and sixteen force calls.

DATA AVAILABILITY

The data that support the findings of this study, including all QE inputs and outputs, force–displacement data, fitted force constants, and the scripts that generate every table and figure, are openly available at `https://github.com/albert-hzbn/SOTC`.

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